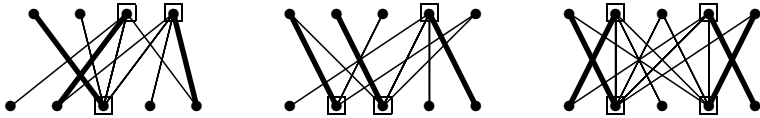


3.MATCHINGS AND FACTORS

3.1. MATCHINGS AND COVERS

3.1.1. Examples of maximum matchings. In each graph below, we show a matching in bold and mark a vertex cover of the same size. We know that the matching has maximum size and the vertex cover has minimum size because the size of every matching in a graph is at most the size of every vertex cover in the graph.



3.1.2. The minimum size of a maximal matching in C_n is $\lceil n/3 \rceil$. Suppose that C_n has a matching of size k . The $n - 2k$ unmatched vertices fall into k “buckets” between matched edges around the cycle. The matching is maximal if and only if none of the buckets contain two vertices. By the pigeonhole principle, there must be some such bucket if $n - 2k \geq k + 1$; hence a matching of size k cannot be maximal if $k \leq (n - 1)/3$.

On the other hand, if $n - 2k \leq k$ and $k \leq n/2$, then we can match two vertices and skip one until we have skipped $n - 2k$, which avoids having two vertices in one bucket. Hence C_n has a maximal matching of size $k \leq n/2$ if and only if $3k \geq n$. We conclude that the minimum size of a maximal matching in C_n is $\lceil n/3 \rceil$.

3.1.3. If $S \subseteq V(G)$ is saturated by some matching in G , then S is saturated by some maximum matching. If M saturates S , then the characterization of maximum matchings implies that a maximum matching M^* can be obtained from M by a sequence of alternating path augmentations. Although edges may be lost in an augmentation, each augmentation continues to saturate all saturated vertices and enlarges the saturated set by 2. Thus S remains saturated in M^* .

When $S \subseteq V(G)$ is saturated by some matching in G , it need not be true that S is saturated by every maximum matching. When G is an odd cycle, all maximum matchings omit distinct vertices.

3.1.4. Let G be a simple graph.

$\alpha(G) = 1$ if and only if G is a complete graph. The independence number is 1 if and only if no two vertices are nonadjacent.

$\alpha'(G) = 1$ if and only if G consists of isolated vertices plus a triangle or nontrivial star. Deleting the vertices of one edge in such a graph leaves no edges remaining. Conversely, suppose that $\alpha'(G) = 1$, and ignore the isolated vertices. Let v be a vertex of maximum degree. If every edge is incident to v , then G is a star. Otherwise, an edge e not incident to v shares an endpoint with every edge incident to v . Since e has only two endpoints, $d(v) = 2$, and there is only one such edge e . Thus G is a triangle.

$\beta(G) = 1$ if and only if G is a nontrivial star plus isolated vertices. The vertex cover number is 1 if and only if one vertex is incident to all edges.

$\beta'(G) = 1$ if and only if $G = K_2$. Since every edge covers two vertices, $\beta'(G) = 1$ requires that $n(G) = 2$, and indeed $\beta'(K_2) = 1$.

3.1.5. $\alpha(G) \geq \frac{n(G)}{\Delta(G)+1}$ for every graph G . Form an independent set S by iteratively selecting a remaining vertex for S and deleting that vertex and all its neighbors. Each step adds one vertex to S and deletes at most $\Delta(G) + 1$ vertices from G . Hence we perform at least $n(G)/(\Delta(G) + 1)$ steps and obtain an independent set at least that big.

3.1.6. If T is a tree with n vertices and independence number k , then $\alpha'(T) = n - k$. The vertices outside a maximum independent set form a vertex cover of size $n - k$. Since trees are bipartite, the König–Egerváry Theorem then applies to yield $\alpha'(T) = \beta(T) = n - \alpha(T) = n - k$.

3.1.7. A graph G is bipartite if and only if $\alpha(H) = \beta'(H)$ for every subgraph H having no isolated vertices. If G is bipartite, then every subgraph H of G is bipartite, and by König’s Theorem the number of edges of H needed to cover $V(H)$ equals $\alpha(H)$ if H has no isolated vertices. If G is not bipartite, then G contains an odd cycle H , and this subgraph H has no isolated vertices and requires $\alpha(H) + 1$ edges to cover its vertices.

3.1.8. Every tree T has at most one perfect matching.

Proof 1 (contradiction). Let M and M' be perfect matchings in a tree. Form the symmetric difference of the edge sets, $M \Delta M'$. Since the matchings are perfect, each vertex has degree 0 or 2 in the symmetric difference, so every component is an isolated vertex or a cycle. Since the tree has no cycle, every vertex must have degree 0 in the symmetric difference, which means that the two matchings are the same.

Proof 2 (induction). For the basis step, a tree with one vertex has no perfect matching; a tree with two vertices has one. For the induction step, consider an arbitrary tree T on $n > 2$ vertices, and consider a leaf v . In any perfect matching, v must be matched to its neighbor u . The remainder of

any matching is a matching in $T - \{u, v\}$. Since each perfect matching in T must contain the edge uv , the number of perfect matchings in T equals the number of perfect matchings in $T - \{u, v\}$.

Each component of $T - \{u, v\}$ is a tree; by the induction hypothesis, each component has at most one perfect matching. The number of perfect matchings in a graph is the product of the number of perfect matchings in each component, so the original T has at most one perfect matching. (More generally, a forest has at most one perfect matching.)

3.1.9. Every maximal matching in a graph G has size at least $\alpha'(G)/2$.

Proof 1 (counting and contrapositive). Let M^* be a maximum matching, and let M be another matching. We show that if $|M| < \alpha'(G)/2$, then M is not a maximal matching. Since M saturates $2|M|$ vertices and $|M| < \alpha'(G)/2$, we conclude that M saturates fewer than $\alpha'(G)$ vertices. This means that M cannot saturate a vertex of every edge of M^* , and there is some edge of M^* that can be added to enlarge M .

Proof 2 (augmenting paths). Let M be a maximal matching, and let M^* be a maximum matching. Consider the symmetric difference $F = M \Delta M^*$. Since the number of edges from M and M^* in a component of F differ by at most one, the symmetric difference contains at least $|M^*| - |M|$ augmenting paths. Since M is maximal, each augmenting path must contain an edge of M (an M -augmenting path of length one is an edge that can be added to M). Thus $|M^*| - |M| \leq |M|$, and we obtain $|M| \geq |M^*|/2 = \alpha'(G)/2$.

Proof 3 (vertex covers). When M is a maximal matching, then the vertices saturated by M form a vertex cover (if an edge had no vertex in this set, then it could be added to M). Since every vertex cover has size at least $\alpha'(G)$, we obtain $2|M| \geq \beta(G) \geq \alpha'(G)$.

3.1.10. If M and N are matchings in a graph G and $|M| > |N|$, then there are matchings M' and N' in G such that $|M'| = |M| - 1$, $|N'| = |N| + 1$, $M' \cap N' = M \cap N$, and $M' \cup N' = M \cup N$. Consider the subgraph F of G consisting of the edges in the symmetric difference $M \Delta N$; this consists of all edges belonging to exactly one of M and N . Since each of M and N is a matching, every vertex has at most one incident edge in M and at most one incident edge in N . Hence the degree of every vertex in F is at most 2.

The components of a graph with maximum degree 2 are paths and cycles. A path or cycle in F alternates edges between M and N . Since $|M| > |N|$, F has a component with more edges of M than of N . Such a component can only be a path P that starts and ends with an edge of M . Form M' from M by replacing $M \cap E(P)$ with $N \cap E(P)$; this reduces the size by one. Form N' from N by replacing $N \cap E(P)$ with $M \cap E(P)$; this increases the size by one. Since we have only switched edges belonging to exactly one of the sets, we have not changed the union or intersection.

3.1.11. If C and C' are cycles in a graph G , then $C \Delta C'$ decomposes into cycles. Since even graphs decompose into cycles (Proposition 1.2.27), it suffices to show that $C \Delta C'$ has even degree at each vertex. The set of edges in a cycle that are incident to v has even size (2 or 0). The symmetric difference of any two sets of even size has even size, since always $|A \Delta B| = |A| + |B| - 2|A \cap B|$.

3.1.12. If C and C' are cycles of length k in a graph with girth k , then $C \Delta C'$ is a single cycle if and only if $C \cap C'$ is a single nontrivial path.

Sufficiency. If $C \cap C'$ is a single path P , then the other paths in C and C' between the endpoints of P share only their endpoints, and hence $C \Delta C'$ is their union and is a single cycle.

Necessity. We know that $C \cap C'$ must have an edge, since otherwise C and C' are edge-disjoint and $C \Delta C'$ has two cycles. Also we may assume that C and C' are distinct, since otherwise $C \Delta C'$ has no edges.

Suppose that P and P' are distinct maximal paths in $C \cap C'$. Now C is the union of four paths P, Q, P', Q' , and C' is the union of four paths P, R, P', R' . Note that Q and Q' may share edges with R and R' . By symmetry, we may assume that P' is no longer than P and that Q and R share the same endpoint of P .

If Q and R also share the same endpoint of P' , then $Q \cup R$ and $Q' \cup R'$ both form closed walks in which (by the maximality of P and P') some edge appears only once. If Q and R do not share the same endpoint of P' , then $P' \cup Q \cup R$ and $P' \cup Q' \cup R'$ both form closed walks in which the edges of P' appear only once. In each case, the two closed walks each contain a cycle, and the sum of their lengths is less than $2k$. This yields a cycle of length less than k in G , which is impossible.

Comment: The statement can fail for longer cycles. In the 3-dimensional cube Q_3 , there are two 6-cycles through antipodal vertices, and their symmetric difference consists of two disjoint 4-cycles.

3.1.13. In an X, Y -bigraph G , if $S \subseteq X$ is saturated by a matching M and $T \subseteq Y$ is saturated by a matching M' , then there is a matching that saturates both S and T . Let F be a subgraph of G with edge set $M \cup M'$. Since each vertex has at most one incident edge from each matching, F has maximum degree 2. Each component of F is an alternating path or an alternating cycle (alternating between M and M'). From a component that is an alternating cycle or an alternating path of odd length, we can choose the edges of M or of M' to saturate all the vertices of the component.

Let P be a component of F that is a path of even length. The edge at one end of P is in M ; the edge at the other end is in M' . Also P starts and ends in the same partite set. If it starts and ends in X , then the ends cannot both be in S , because only one endpoint of P is saturated by M .

Choosing the edges of M in this component will thus saturate all vertices of S and T contained in $V(P)$. Similarly, choosing the edges of M' from any component of F that is a path of even length with endpoints in Y will saturate all the vertices of $S \cup T$ in that component.

3.1.14. Matchings in the Petersen graph.

Deleting any perfect matching leaves $C_5 + C_5$. Deleting a perfect matching leaves a 2-regular spanning subgraph, which is a disjoint union of cycles. Since the Petersen graph has girth 5, the only possible coverings of the vertices by disjoint cycles are $C_5 + C_5$ and C_{10} .

If a 10-cycle exists, with vertices $[v_1, \dots, v_{10}]$ in order, then the remaining matching consists of chords. Two consecutive vertices cannot neighbor their opposite vertices on the cycle, since that creates a 4-cycle. Similarly, the neighbors must be at least four steps away on the cycle. Hence we may assume by symmetry that $v_1 \leftrightarrow v_5$. Now making v_{10} adjacent to any of $\{v_6, v_5, v_4\}$ creates a cycle of length at most 4, so there is no way to insert the remaining edges.

a) *The Petersen graph has twelve 5-cycles.* Each edge extends to P_4 in four ways by picking an incident edge at each endpoint. Since the graph has diameter two and girth 5, every P_4 belongs to exactly one 5-cycle through an additional vertex. Since there are 15 edges, we have generated 60 5-cycles, but each 5-cycle is generated five times.

b) *The Petersen graph has six perfect matchings.* Since the Petersen graph has girth five, the five remaining edges incident to any 5-cycle form a perfect matching, and deleting them leaves a 5-cycle on the complementary vertices. Hence the 5-cycles group into pairs of 5-cycles with a matching between them. Since every matching leaves $C_5 + C_5$, every matching arises in this way, and by part (b) there are six of them.

3.1.15. Matchings in k -dimensional cubes.

a) *For $k \geq 2$, if M is a perfect matching of Q_k , then there are an even number of edges in M whose endpoints differ in coordinate i .* Let V_0 and V_1 be the sets of vertices having 0 and 1 in coordinate i , respectively. Each has even size. Since the vertices of V_r not matched to V_{1-r} must be matched within V_r , the number of vertices matched by edges to V_{1-r} must be even.

b) *Q_3 has nine perfect matchings.* There are four edges in each such matching, with an even number distributed to each coordinate. The possible distributions are $(4, 0, 0)$ and $(2, 2, 0)$. There are three matchings of the first type. For the second type, we pick a direction to avoid crossing, pick one of the two matchings in one of the 4-cycles, and then the choice of the matching in the other 4-cycle is forced to avoid making all four edges change the same coordinate. Hence there are $3 \cdot 2 \cdot 1$ perfect matchings of the second type.

3.1.16. *When $k \geq 2$, the k -dimensional hypercube Q_k has at least $2^{(2^{k-2})}$ perfect matchings.*

Proof 1 (induction on k). Let m_k denote the number of perfect matchings. Note that $m_2 = 2$, which satisfies the inequality. When $k > 2$, we can choose matchings independently in each of two disjoint subcubes of dimension $k - 1$. The number of such matchings is m_{k-1}^2 . By the induction hypothesis, this is at least $(2^{2^{k-3}})^2$, which equals $2^{2^{k-2}}$.

Comment: Since we could choose the two disjoint subcubes in k ways, we can recursively form km_{k-1}^2 perfect matchings in this way, some of which are counted more than once.

Proof 2 (direct construction). Pick two coordinates. There are 2^{k-2} copies of Q_2 in which those two coordinates vary, and two choices of a perfect matching in each copy of Q_2 . This yields $2^{2^{k-2}}$ perfect matchings. (Since we can choose the two coordinates in $\binom{k}{2}$ ways, we can generate $\binom{k}{2} 2^{2^{k-2}}$ perfect matchings, but there is some repetition.)

3.1.17. *In every perfect matching in the hypercube Q_k , there are exactly $\binom{k-1}{i}$ edges that match vertices with weight i to vertices with weight $i + 1$, where the weight of a vertex is the number of 1s in its binary k -tuple name.*

Proof 1 (induction on i). Since the vertex of weight 0 must match to a vertex of weight 1, the claim holds when $i = 0$. For the induction step, the induction hypothesis yields $\binom{k-1}{i-1}$ vertices of weight $i - 1$ matched to vertices of weight i . The remaining vertices of weight i must match to vertices of weight $i + 1$. Since $\binom{k}{i} - \binom{k-1}{i-1} = \binom{k-1}{i}$, the claim follows.

Proof 2 (canonical forms). Let M^* be the matching where every edge matches vertices with 0 and 1 in the last coordinate. The number of edges matching weight i to weight $i + 1$ is the number of choices of i ones from the first $k - 1$ positions, which is $\binom{k-1}{i}$.

It now suffices to prove that every perfect matching M has the same weight distribution as M^* . The symmetric difference of M and M^* is a union of even cycles alternating between M and M^* , plus isolated vertices saturated by the same edge in both matchings. It suffices to show that the weight distribution on each cycle is the same for both matchings.

The edges joining vertices of weights i and $i + 1$ along a cycle C alternate appearing with increasing weight and with decreasing weight, since weight changes by 1 along each edge. For the same reason, the number of edges along C from a vertex to the next appearance of a vertex with the same weight is even. Since C alternates between M and M^* , this means that the edges joining vertices of weights i and $i + 1$ alternate between M and M^* . Hence there is the same number of each type, as desired.

3.1.18. *The game of choosing adjacent vertices, where the last move wins. Suppose that G has a perfect matching M . Whenever the first player*

chooses a vertex, the second player takes its mate in M . This vertex is available, because after each move of the second player the set of vertices visited forms a set of full edges in M , and the first player cannot take two vertices at a time. Thus with this strategy, the second player can always make a move after any move of the first player and never loses.

If G has no perfect matching, then let M be a maximum matching in G . The first player starts by choosing a vertex not covered by M . Thereafter, whenever the second player chooses a vertex x , the first player chooses the mate of x in M . The vertex x must be covered by M , else x completes an M -augmenting path using all the vertices chosen thus far. Thus the first player always has a move available and does not lose.

3.1.19. A family $A_1, \dots, A_m$ of subsets of Y has a system of distinct representatives if and only if $|\bigcup_{j \in S} A_j| \geq |S|$ for every $S \subseteq [m]$. Form an X, Y -bigraph G with $X = \{1, \dots, m\}$ and $Y = \{y_1, \dots, y_n\}$. Include the edge iy_j if and only if $y_j \in A_i$. A set of edges in G is a matching if and only if its endpoints in Y form a system of distinct representatives for the sets indexed by its endpoints in X . The family has a system of distinct representatives if and only if G has a matching that saturates X .

It thus suffices to show that the given condition is equivalent to Hall's condition for saturating X . If $S \subseteq X$, then $N_G(S) = \bigcup_{i \in S} A_i$, so $|N(S)| \geq |S|$ if and only if $|\bigcup_{i \in S} A_i| \geq |S|$.

3.1.20. An extension of Hall's Theorem using stars with more than two vertices. We form an X, Y -bigraph G with partite sets $X = x_1, \dots, x_n$ for the trips and $Y = y_1, \dots, y_m$ for the people, and edge set $\{x_i y_j : \text{person } j \text{ likes trip } i\}$. To fill each trip to its capacity c_i , we seek a subgraph whose components are stars, with degree c_i at x_i .

Form an X', Y -bigraph G' by making n_i copies of each vertex x_i . Now G has the desired stars if and only if G' has a matching that saturates X' . Thus the desired condition for G should become Hall's Condition for G' .

In G' , the neighborhoods of the copies of a vertex of x are the same. Hence Hall's Condition will hold if and only if it holds whenever $S \subseteq X'$ consists of all copies of each vertex of X for which it includes any copies. That is, Hall's Condition reduces to requiring $|N(T)| \geq \sum_{x_i \in T} c_i$ for all $T \subseteq X$. This condition is necessary, since the trips in T need this many distinct people. It is sufficient, because it implies Hall's Condition for G' .

3.1.21. If G is an X, Y -bigraph such that $|N(S)| > |S|$ whenever $\emptyset \neq S \subset X$, then every edge of G belongs to some matching that saturates X . Let xy be an edge of G , with $x \in X$ and $y \in Y$, and let $G' = G - x - y$. Each set $S \subseteq X - \{x\}$ loses at most one neighbor when y is deleted. Combining this with the hypothesis yields $|N_{G'}(S)| \geq |N_G(S)| - 1 \geq |S|$. Thus G' satisfies

Hall's Condition and has a matching that saturates $X - \{x\}$. With the edge xy , this completes a matching in G that contains xy and saturates X .

3.1.22. A bipartite graph G has a perfect matching if and only if $|N(S)| \geq |S|$ for all $S \subseteq V(G)$. This conclusion does not hold for non-bipartite graphs. In an odd cycle, we obtain neighbors for a set of vertices by taking the vertices immediately following them on the cycle. Thus $|N(S)| \geq |S|$ for all $S \subseteq V$, but the graph has no perfect matching. Complete graphs of odd order also form counterexamples. For bipartite graphs, we give two proofs.

Proof 1 (graph transformation). Let G' be a bipartite graph consisting of two disjoint copies of G , where each partite set in G' consists of one copy of X and one copy of Y ; call these X' and Y' . Then G' has a perfect matching if and only if G has a perfect matching. Since $|X'| = |Y'|$, G' has a perfect matching if and only if it has a matching that completely saturates X' .

By Hall's Theorem, G' has a matching saturating X' if and only if $|N(S')| \geq |S'|$ for all $S' \subseteq X'$. Given $S' \subseteq X'$, let $T_1 = S' \cap X$ and $T_2 = S' - T_1$. Let $S \subseteq V(G)$ be the set of vertices consisting of T_1 in X plus the vertices of Y having copies in T_2 . This establishes a bijection between subsets S' of X' and subsets S of $V(G)$, with $|S'| = |S|$. Also $|N(S')| = |N(S)|$, by the construction of G' .

Hence Hall's condition is satisfied for G' if and only if the condition of this problem holds in G . In summary, we have shown

$$[G \text{ has a 1-factor}] \Leftrightarrow G' \text{ has a 1-factor} \Leftrightarrow$$

$$|N(S')| \geq |S'| \text{ for all } S' \subseteq X' \Leftrightarrow |N(S)| \geq |S| \text{ for all } S \subseteq V(G).$$

Proof 2 (by Hall's Theorem). *Necessity:* Let M be a perfect matching, and let S be a subset of $V(G)$. Vertices of S are matched to distinct vertices of $N(S)$ by M , so $|N(S)| \geq |S|$. *Sufficiency:* If $|N(S)| \geq |S|$ for all $S \subseteq V$, then $|N(A)| \geq |A|$ for all $A \subseteq X$. By Hall's Theorem, the graph thus has a matching M that saturates X . Thus $|Y| \geq |X|$, and the condition $|N(Y)| \geq |Y|$ yields $|X| \geq |Y|$. Thus $|Y| = |X|$, and M is a perfect matching.

3.1.23. Alternative proof of Hall's Theorem. Given an X, Y -bigraph G , we prove that Hall's Condition suffices for a matching that saturates X . Let $m = |X|$. For $m = 1$, the statement is immediate.

Induction step: $m > 1$. If $|N(S)| > |S|$ for every nonempty proper subset $S \subset X$, select any neighbor y of any vertex $x \in X$. Deleting y reduces the size of the neighborhood of each subset of $X - \{x\}$ by at most 1. Hence Hall's Condition holds in $G' = G - x - y$. By the induction hypothesis, G' has a matching that saturates $X - \{x\}$, which combines with xy to form a matching that saturates X .

Otherwise, $|N(S)| = |S|$ for some nonempty proper subset $S \subset X$. Let $G_1 = G[S \cup N(S)]$, and let $G_2 = G - V(G_1)$. Because the neighbors of

vertices in S are confined to $N(S)$, Hall's Condition for G implies Hall's Condition for G_1 . For G_2 , consider $T \subseteq X - S$. Since $|N_G(T \cup S)| \geq |T \cup S|$, we obtain

$$N_{G_2}(T) = N_G(T \cup S) - N_G(S) \geq |T \cup S| - |S| = |T|.$$

Thus Hall's Condition holds for both G_1 and G_2 . By the induction hypothesis, G_1 has a matching that saturates S , and G_2 has a matching that saturates $X - S$. Together, they form a matching that saturates X .

3.1.24. A square matrix of nonnegative integers is a sum of k permutation matrices if and only if each row and column sums to k . If A is the sum of k permutation matrices, then each matrix adds one to the sum in each row and column, and each row or column of A has sum k .

For the converse, let A be a square matrix with rows and columns summing to k . We use induction on k to express A as a sum of k permutation matrices. For $k = 1$, A is a permutation matrix.

For $k > 1$, form a bipartite graph G with vertices $x_1, \dots, x_n$ and $y_1, \dots, y_n$ so that the number of edges joining x_i and y_j is $a_{i,j}$. The graph G is bipartite and regular, so by the Marriage Theorem it has a perfect matching. Let $b_{i,j} = 1$ if $x_i y_j$ belongs to this matching and $b_{i,j} = 0$ otherwise; the resulting matrix B is a permutation matrix. Each row and column of B has exactly one 1. Thus $A' = B - A$ is a nonnegative integer matrix whose rows and columns sum to $k - 1$. Applying the induction hypothesis to A' yields $k - 1$ additional permutation matrices that with B sum to A .

3.1.25. A nonnegative doubly stochastic matrix can be expressed as a convex combination of permutation matrices. For simplicity, we allow multiples of doubly stochastic matrices and prove a superficially more general statement. We use induction on the number of nonzero entries to prove that if Q is a matrix of nonnegative entries in which every row and every column sums to t , then Q can be expressed as a linear combination of permutation matrices with nonnegative coefficients summing to t .

If Q has exactly n nonzero entries, then Q is t times a permutation matrix, because Q must have at least one nonzero entry in every row and column. If Q has more nonzero entries, begin by defining a bipartite graph G with $x_i \leftrightarrow y_j$ if and only if $Q_{i,j} > 0$. If G has a perfect matching, then the edges $x_i y_{\sigma(i)}$ of the matching correspond to a permutation σ with permutation matrix P .

Let ε be the minimum (positive) value in the positions of Q corresponding to the 1's in P . The matrix $Q' = Q - \varepsilon P$ is a nonnegative matrix with fewer nonzero entries than Q , and row and column sums $t - \varepsilon$. By the induction hypothesis, we can express Q' as a nonnegative combination

$\sum_{i=1}^m c_i P_i$, with $\sum_{i=1}^m c_i = t - \varepsilon$. Hence $Q = \sum c_i P_i + \varepsilon P$. With $c_{m+1} = \varepsilon$ and $P_{m+1} = P$, we have expressed Q in the desired form.

It remains to prove that G has a perfect matching; we show that it satisfies Hall's condition. If S is a subset of X corresponding to a particular set of rows in Q , we need only show that these rows have nonzero entries in at least $|S|$ columns altogether. This follows because the total nonzero amount in the rows S is $t|S|$. Since each column contains only a total of t , it is not possible to contain a total of $t|S|$ in fewer than $|S|$ columns.

Comment. When the entries of Q are rational, the result follows directly from the Marriage Theorem. Multiplying Q by the least common denominator d of its positive entries converts it to an integer matrix in which all rows and columns sum to d . The entry in position i, j now is the number of edges joining x_i and y_j in a d -regular bipartite graph (multiple edges allowed). The Marriage Theorem implies that the graph has a perfect matching. By induction on d , it can be decomposed into perfect matchings. These matchings correspond to permutation matrices. In the expression of Q as a convex combination of these matrices, we give weight a/d to a permutation matrix arising a times in the list of matchings.

3.1.26. Achieving columns with all suits. The cards in an n by m array have m values and n suits, with each value on one card in each suit.

a) It is always possible to find a set of m cards, one in each column, having the m different values. Form a X, Y -bigraph in which X represents the columns and Y represents the values, with r edges from $x \in X$ to $y \in Y$ if value y appears r times in column x . Since each column contains n cards and each value appears in n positions (once in each suit), the multigraph is n -regular. By the Marriage Corollary to Hall's Theorem, every nontrivial regular bipartite graph has a perfect matching. (This applies also when multiple edges are present, which can occur here.) A perfect matching selects m distinct values occurring in the m columns.

(Using Hall's Theorem directly, a set S of k columns contains nk cards. Since there are n cards of each value, S contains cards of at least k values. Hence the graph satisfies Hall's condition and has a perfect matching.)

b) By a sequence of exchanges of cards of the same value, the cards can be rearranged so that each column consists of n cards of distinct suits. Making each column consist of n cards of different suits is equivalent to spreading each suit across all columns. The full result follows by induction on n , with $n = 1$ as a trivial basis step.

For the induction step, when $n > 1$, use part (a) to find cards of distinct values representing the m columns. Then perform at most one exchange for each value to bring the values in a single suit to those positions. Positions within a column are unimportant, so we can treat the other suits as an

instance of the problem with $n - 1$ suits. We apply the induction hypothesis to fix up the remaining suits.

The problem can always be solved using at most $mn - \sum_{k \leq n} \lceil m/k \rceil$ exchanges. The worst case requires at least $\lfloor m/n \rfloor n(n - 1)/2$ exchanges.

3.1.27. *The second player can force a draw in a positional game if $a \geq 2b$, where a is the minimum size of a winning set and b is the maximum number of winning sets containing a particular position. Let P be the set of positions, and let $W_1, \dots, W_m$ be the winning sets of positions. With $|P| = n$, let G be the bipartite graph on $n + 2m$ vertices with partite sets $P = \{p_1, \dots, p_n\}$ and $W = \{w_1, \dots, w_m\} \cup \{w'_1, \dots, w'_m\}$ by creating two edges $p_i w_j$ and $p_i w'_j$ for each incidence $p_i \in W_j$ of a position in a winning set.*

If G has a matching M that saturates W , then Player 2 can use M to force a draw. When the position taken by Player 1 on a given move is matched to one of $\{w_i, w'_i\}$ in M , Player 2 responds by taking the position matched to the other one of these two elements. Player 1 thus can never obtain all the positions of a winning set. (If Player 1 takes an unmatched position, Player 2 can respond by taking any available position.)

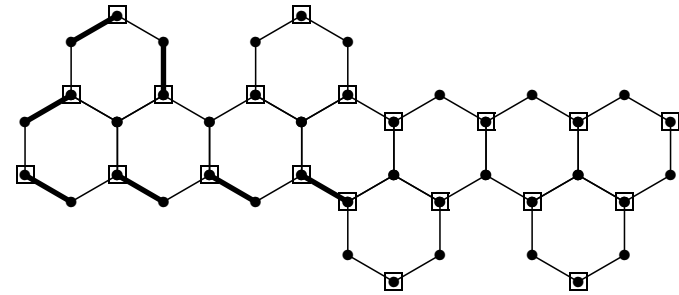
The existence of such a matching follows from $a \geq 2b$ by Hall's condition. If $S \subseteq W$, then S has representatives (w or w') of at least $|S|/2$ winning sets. Since each position appears in at most b winning sets, the number of positions in the union of these winning sets is at least $a(|S|/2)/b \geq |S|$. Thus $|N(S)| \geq |S|$ for every $S \subseteq W$.

3.1.28. *A graph with no perfect matching.*

Proof 1 (vertex cover). The graph has 42 vertices, so a perfect matching would have 21 edges. The marked vertices form a vertex cover of size 20. The edges of a matching must be covered by distinct vertices in a vertex cover, so there is no matching with more than 20 edges.

Proof 2 (Hall's condition). Using two labels X and Y , we obtain a bipartition of the graph. Partite set X consists of the marked vertices in the left half of the picture and the unmarked vertices in the right half. This is an independent set of size 21, and the remaining vertices also form an independent set Y of size 21.

Hall's Condition is a necessary condition for a perfect matching; we show that Hall's Condition does not hold. Let S be the subset of X consisting of the 11 unmarked vertices in the right half of the graph. The neighbors of vertices in S are the 10 marked vertices in the right half of the graph. Thus $|N(S)| < |S|$.



Proof 3 (other dual problems). In every graph $\alpha'(G) + \beta'(G) = n(G)$, so it suffices to show that at least 22 edges are needed to cover $V(G)$. Also $\beta'(G) \geq \alpha(G)$ always, since distinct edges are needed to cover the vertices of an independent set. Thus it suffices to show that G has an independent set of size at least 22. Such a set is given by the unmarked vertices above (the complement of a vertex cover).

Proof 4 (augmenting paths). Having found a matching M of size 20, one can prove that there is no perfect matching by following all possible M -alternating paths from one M -unsaturated vertex to show that none reaches the other unsaturated vertex. In this particular example, this method is not too difficult.

Proof 5 (symmetry and case analysis). The graph has two edges whose deletion leaves two isomorphic components of order 21. Since 21 is odd, a perfect matching must use exactly one of the two connecting edges. By symmetry, we may assume it is the one in bold above. This forces a neighboring vertex of degree 2 to be matched to its other neighbor, introducing other bold edge. Repeating this argument yields a path of bold edges forced into the matching. As soon as this leaves a vertex with no available neighbor, we have proved that a perfect matching cannot be completed.

3.1.29. *Every bipartite graph G has a matching of size at least $e(G)/\Delta(G)$. Each vertex of G covers at most $\Delta(G)$ edges. Since all edges must be covered in a vertex cover, this yields $\beta(G) \geq e(G)/\Delta(G)$. By the König–Egerváry Theorem, $\alpha'(G) = \beta(G)$ when G is bipartite. Thus $\alpha'(G) \geq e(G)/\Delta(G)$.*

Every subgraph of $K_{n,n}$ with more than $(k - 1)n$ edges has a matching of size at least k . Such a graph G is a simple bipartite graph with partite sets of size n . Thus $\Delta(G) \leq n$, and we compute $\alpha'(G) \geq e(G)/\Delta(G) > (k - 1)n/n = k - 1$. Thus G has a matching of size k .

3.1.30. *The maximum number of edges in a simple bipartite graph that has no matching with k edges and no star with l edges is $(k - 1)(l - 1)$. If G is a bipartite graph having no matching with k edges, then G has a vertex cover using at most $k - 1$ vertices. If G is a simple graph having no star*

with l edges, then each vertex covers at most $l - 1$ edges. Hence the vertex cover covers at most $(k - 1)(l - 1)$ edges, which must be all the edges of G . The bound is achieved by $(k - 1)K_{1,l-1}$.

3.1.31. Hall's Theorem from the König–Egerváry Theorem. By the König–Egerváry Theorem, an X, Y -bigraph G fails to have a matching that saturates X if and only if G has a vertex cover of size less than $|X|$. Let Q be such a cover, with $R = Q \cap X$ and $T = Q \cap Y$. Because Q is a vertex cover, there is no edge from $X - R$ to $Y - T$, which means that $N(X - R) \subseteq T$. This yields

$$|N(X - R)| \leq |T| = |Q| - |R| < |X| - |R| = |X - R|.$$

We have used the König–Egerváry Theorem to show that absence of a matching that saturates X yields a violation of Hall's Condition. Thus Hall's Condition is sufficient for such a matching.

Similarly, if $|N(S)| < |S|$ for some $S \subseteq X$, then $N(S) \cup X - S$ is a vertex cover of size less than $|X|$, and there is no matching of size $|X|$. Hence Hall's Condition also is necessary.

3.1.32. If G is a bipartite graph with partite sets X, Y , then $\alpha'(G) = |X| - \max_{S \subseteq X} (|S| - |N(S)|)$. Let $d = \max_{S \subseteq X} (|S| - |N(S)|)$. The case $S = \emptyset$ implies that $d \geq 0$. Choose $T \subseteq X$ such that $|T| - |N(T)| = d$. Because saturated vertices of T must have distinct neighbors in any matching and only $|T| - d$ neighbors are available, every matching leaves at least d vertices (of T) unsaturated. Thus $\alpha'(G) \leq |X| - d$.

To prove that G has a matching as large as $|X| - d$, we form a new graph G' by adding d vertices to the partite set Y and making all of them adjacent to all of X . This adds d vertices to $N(S)$ for each $S \subseteq X$, which yields $|N_{G'}(S)| \geq |S|$ for all $S \subseteq X$. By Hall's Theorem, G' has a matching saturating all of X . When we delete the new vertices of G' , we lose at most d edges of the matching. Hence what remains is a matching of size at least $|X| - d$ in G , as desired.

3.1.33. König–Egerváry from Exercise 3.1.32. Always $\alpha'(G) \leq \beta(G)$, so it suffices to show that a bipartite graph G has a matching and a vertex cover of the same size. Consider an X, Y -bigraph G in which S is a subset of X with maximum deficiency. By part (a), $\alpha'(G) = |X| - |S| + |N(S)|$.

Let $R = (X - S) \cup (N(S))$. By the definition of $N(S)$, there are no edges joining S and $Y - N(S)$. Therefore, R is a vertex cover of G . The size of R is $|X| - |S| + |N(S)|$, which equals $\alpha'(G)$. Thus G has a matching and a vertex cover of the same size, as desired.

3.1.34. When G is an X, Y -bigraph with no isolated vertices and the deficiency of a set S is $|S| - |N(S)|$, the graph G has a matching that saturates X

if and only if each subset of Y has deficiency at most $|Y| - |X|$. Using Hall's Theorem, it suffices to show that $|N(S)| \geq |S|$ for all $S \subseteq X$ if and only if $|T| - |N(T)| \leq |Y| - |X|$ for all $T \subseteq Y$.

We rewrite the latter condition as $|X| - |N(T)| \leq |Y| - |T|$. Since every vertex of $X - N(T)$ has no neighbor in T , we have $N(X - N(T)) \subseteq Y - T$. If Hall's Condition holds, then applying it with $S = X - N(T)$ yields $|X| - |N(T)| = |S| \leq |N(S)| \leq |Y| - |T|$, which is the desired condition.

Conversely, suppose that $|X| - |N(T)| \leq |Y| - |T|$ for all $T \subseteq Y$. Given $S \subseteq X$, let $T = Y - N(S)$. Since T omits all neighbors of vertices in S , we have $S \subseteq X - N(T)$. Now $|S| \leq |X| - |N(T)| \leq |Y| - |T| = |N(S)|$. Hence Hall's Condition holds.

3.1.35. A bipartite graph G in $K_{X,Y}$ fails to have $(k + 1)K_2$ as an induced subgraph if and only if each $S \subseteq X$ has a subset of size at most k with neighborhood $N(S)$. For any $S \subseteq X$, let T be a minimal subset of S with neighborhood $N(S)$. By the minimality of T , G has an induced matching of size $|T|$. Hence if G has no induced matching of size k , each $S \subseteq X$ has a subset of size at most k with neighborhood $N(S)$. Conversely, if $k + 1K_2$ does occur as an induced subgraph, then the set of its vertices in X have no subset of size at most k with the same neighborhood.

3.1.36. If a bipartite graph G has a matching saturating a partite set X of size m , then at most $\binom{m}{2}$ edges of G belong to no matching of size m . There are at least three distinguishable ways to get the bound. The most direct one, and the one that suggests the extremal graph proving that no smaller bound is possible, considers pairs of vertices.

Proof 1. First note that, after renumbering the vertices so the edges of the given matching are $\{x_i y_i\}$, every edge involving any other vertex of Y belongs to a perfect matching, so it suffices to restrict attention to the subgraph induced by X and $y_1, \dots, y_m$. Consider the edges $x_i y_j$ and $x_j y_i$, for $j < i \leq m$. If both edges are present, then neither belongs to no maximum matching, since they can be exchanged for $x_i y_i$ and $x_j y_j$ in the original matching. Taking one from each such pair bounds the number of unmatchable edges by $\binom{m}{2}$. On the other hand, taking all $x_i y_j$ with $j < i \leq m$ yields a graph in which $\{x_i y_i\}$ is the unique maximum matching. If any edge $x_i y_i$ is deleted from this graph, then $|\text{Adj}(\{x_1, \dots, x_i\})| = i - 1$.

Proof 2. Induction on m . As before, reduce attention to the edges between X and $y_1, \dots, y_m$. Among X , let x be the vertex of maximum degree in this subgraph. It has degree at most m , and its deletion leaves a graph satisfying the hypotheses for $m - 1$. Allowing for the matched edge involving x , this gives a bound of $\binom{m-1}{2} + m - 1 = \binom{m}{2}$ on the unmatchable edges. Note that x corresponds to x_m in the example above.

Proof 3. Let S be an arbitrary subset of X . If $|N(S)| > |S|$ for ev-

ery proper nonempty subset S of X , then every edge belongs to a perfect matching. To show this, delete the endpoints of an edge xy . This reduces the adjacency set of any S not containing x by at most 1, so the reduced graph has a perfect matching, and replacing xy yields a perfect matching of the original graph containing xy . So, assume there is an S with $|N(S)| = |S| = s$, and let $N(S) = T$. Then the subgraphs induced by $S \cup T$, $(X - S) \cup T$, and $(X - S) \cup (Y - T)$ partition the edges of G . The first and last have perfect matchings, and an edge there fails to appear in a perfect matching of G if and only if it appears in no perfect matching of the subgraph. No edge of the middle graph appears in a perfect matching of G . By induction and the fact that $|T| = |S|$, the bound on the number of edges that appear in no perfect matching of G is $\binom{s}{2} + s(m - s) + \binom{m-s}{2} = \binom{m}{2}$.

3.1.37. Let G be an X, Y -bigraph having a matching that saturates X .

a) If $S, T \subseteq X$ are sets such that $|N(S)| = |S|$ and $|N(T)| = |T|$, then $|N(S \cap T)| = |S \cap T|$.

Proof 1 (manipulation of sets). Since G has a matching that saturates X , $|N(S \cap T)| \geq |S \cap T|$, and $|N(S \cup T)| \geq |S \cup T|$. Also $N(S \cup T) = N(S) \cup N(T)$ and $N(S \cap T) \subseteq N(S) \cap N(T)$. Together, these statements yield

$$|S \cup T| + |S \cap T| \leq |N(S \cup T)| + |N(S \cap T)| \leq |N(S) \cup N(T)| + |N(S) \cap N(T)| \\ = |N(S)| + |N(T)| = |S| + |T|$$

Since the two ends of this string of expressions are equal, the inequalities along the way hold with equality. In particular, $|N(S \cap T)| = |S \cap T|$ and $|N(S \cup T)| = |S \cup T|$.

Proof 2 (characterization of sets with no excess neighbors). If M is a matching that saturates X , then $|N(S)| = |S|$ if and only if every vertex of $N(S)$ is matched by M to a vertex of S . If $|N(S)| = |S|$ and $|N(T)| = |T|$, then every vertex of $N(S)$ is matched into S and every vertex of $N(T)$ is matched into T . Since $N(S \cap T) \subseteq N(S) \cap N(T)$, we conclude that every vertex of $N(S \cap T)$ is matched into $S \cap T$, and therefore $|N(S \cap T)| = |S \cap T|$.

b) There is a vertex $x \in X$ such that every edge incident to x belongs to some matching that saturates X . We use induction on $|X|$. If $|N(S)| > |S|$ for every nonempty proper subset of X , then Hall's Condition holds for the graph obtained by deleting the endpoints of any edge. Thus each edge can be combined with a matching saturating what remains of X in the graph obtained by deleting its endpoints, so every edge of G belongs to some matching saturating X .

In the remaining case, there is a nonempty proper subset $S \subseteq X$ such that $|N(S)| = |S|$. Let G_1 be the subgraph of G induced by $S \cup N(S)$, and let G_2 be the subgraph obtained by deleting $S \cup N(S)$. As in the proof of Hall's Theorem, the graph G_2 obtained by deleting $S \cup N(S)$ satisfies Hall's

Condition (the proof of $|N_{G_2}(T)| \geq |T|$ follows from $|N_G(T \cup S)| \geq |T \cup S|$). Thus G_2 has a matching saturating $X - S$.

The subgraph G_1 also satisfies Hall's condition, since it retains all neighbors of each vertex of S . By the induction hypothesis, S has a vertex x such that every edge incident to x belongs to a matching in G_1 that saturates S . These matchings can be combined with a single matching in G_2 that saturates $X - S$ to obtain matchings in G that saturate X . Hence the vertex x serves as the desired vertex in G .

It appears that part (a) is not needed to solve part (b).

3.1.38. Pairing up farms and hunting ranges. Suppose the unit of area is the size of one range. Let G be the bipartite graph between hunting ranges and farms formed by placing an edge between a hunting range and a farm if the area of their intersection is at least ε , where $\varepsilon = 4/(n+1)^2$ if n is odd and $\varepsilon = 4/[n(n+2)]$ if n is even. We prove that this graph has a perfect matching, which yields the desired assignments.

Let H be the union of some set of k hunting ranges. Let $f_1 \geq \dots \geq f_n$ be the areas of intersection with H of the farms, and let F be the set of k farms having largest intersection with H . If $f_k = \alpha$, then the area of H is bounded by $\sum_{i=1}^{k-1} g_i + \alpha(n+1-k) \leq k-1 + \alpha(n+1-k)$. It also equals k , so we have $\alpha \geq 1/(n+1-k)$. Since we have k farms meeting H with area at least $1/(n+1-k)$, we find for each farm in F a hunting range contained in H that intersects the farm with area at least $1/[k(n+1-k)] \geq \varepsilon$. Hence any set of k hunting ranges has at least k neighbors in G , which guarantees the matching.

Note that ε is the largest possible guaranteed minimum intersection. Let $k = \lceil n/2 \rceil$. With hunting ranges in equal strips, we can arrange that some set of $k-1$ farms intersects each of the first k hunting ranges with area $1/k$, and the remaining farms intersect each of the first k hunting ranges with area ε , since $(k-1)/k + (n+1-k)\varepsilon = 1$. Now one of the first k hunting ranges must be matched with area ε .

3.1.39. $\alpha(G) \leq n(G) - e(G)/\Delta(G)$. Let S be an independent set of size $\alpha(G)$. Since $V(G) - S$ is a vertex cover, summing the vertex degrees in $V(G) - S$ provides an upper bound on $e(G)$. Thus $e(G) \leq (n(G) - \alpha(G))\Delta(G)$, which is equivalent to the desired inequality.

If G is regular, then $\alpha(G) \leq n(G)/2$. In the previous inequality, set $e(G) = n(G)\Delta(G)/2$.

3.1.40. If G is a bipartite graph, then $\alpha(G) = n(G)/2$ if and only if G has a perfect matching. Since $\alpha(G) = n(G) - \beta(G) = n(G) - \alpha'(G)$ by Lemma 3.1.21 and the König-Egerváry Theorem, we have $\alpha(G) = n(G)/2$ if and only if $\alpha'(G) = n(G)/2$.

3.1.41. (corrected statement) *If G is a nonbipartite n -vertex graph with exactly one cycle C , then $\alpha(G) \geq (n-1)/2$, with equality if and only if $G - V(C)$ has a perfect matching.* The cycle C must have odd length, say k . Let e be an edge of C , and let $G' = G - e$. The graph G' is bipartite, so $\alpha(G - e) \geq n/2$. An independent set S in $G - e$ is also independent in G unless it contains both endpoints of e . If $|S| > n/2$, then we can afford to drop one of these vertices. If $|S| = n/2$, then we can take the other partite set instead to avoid the endpoints of e . In each case, $\alpha(G) \geq (n-1)/2$.

If $G - V(C)$ has a perfect matching, then an independent set is limited to $(k-1)/2$ vertices of C and $(n-k)/2$ vertices outside C , so $\alpha(G) \leq (n-1)/2$ and equality holds.

For the converse, observe that deleting $E(C)$ leaves a forest F in which each component has a vertex of G . Let H be a component of F , with x being its vertex on C , and let r be its order. If $H - x$ has no perfect matching, then $\alpha'(H - x) \leq r/2 - 1$ (that is, it cannot equal $(r-1)/2$). Now $\beta(H - x) \leq r/2 - 1$, by König-Egerváry, and $\alpha(H - x) \geq r/2$, since the complement of a vertex cover is an independent set. Since this independent set does not use x , we can combine it with an independent set of size at least $(n-r)/2$ in the bipartite graph $G - V(H)$ to obtain $\alpha(G) \geq n/2$. Since this holds for each component of F , $\alpha(G) = (n-1)/2$ requires a perfect matching in $G - V(C)$. (This direction can also be proved by induction on $n - k$.)

3.1.42. *The greedy algorithm produces an independent set of size at least $\sum_{v \in V(G)} \frac{1}{d(v)+1}$ in a graph G .* The algorithm iteratively selects a vertex of minimum degree in the remaining graph and deletes it and its neighbors. We prove the desired bound by induction on the number of vertices.

Basis step: $n = 0$. When there are no vertices, there is no contribution to the independent set, and the empty sum is also 0.

Induction step $n > 0$. Let x be a vertex of minimum degree, let $S = \{x\} \cup N(x)$, and let $G' = G - S$. The algorithm selects x and then seeks an independent set in G' . We apply the induction hypothesis to G' to obtain a lower bound on the contribution that the algorithm obtains from G' . Thus the size of the independent set found in G is at least $1 + \sum_{v \in V(G')} \frac{1}{d_{G'}(v)+1}$.

Note that $\{x\} \cup N(x)$ is a set of size $d_G(x) + 1$, and the choice of x as a vertex of minimum degree implies that each vertex in this set contributes at most $d_G(x) + 1$ to the desired sum. Thus

$$\begin{aligned} 1 + \sum_{v \in V(G')} \frac{1}{d_{G'}(v) + 1} &= \sum_{v \in S} \frac{1}{d_G(x) + 1} + \sum_{v \in V(G) - S} \frac{1}{d_{G-S}(v) + 1} \\ &\geq \sum_{v \in S} \frac{1}{d_G(v) + 1} + \sum_{v \in V(G) - S} \frac{1}{d_G(v) + 1}. \end{aligned}$$

Thus the algorithm finds an independent set at least as large as desired.

3.1.43. *Consequences of Gallai's Theorem (G has no isolated vertices).*

a) *A maximal matching M is a maximum matching if and only if it is contained in a minimum edge cover.* If M is a maximal matching, then the smallest edge cover L containing M adds one edge to cover each M -unsaturated vertex, since no edge covers two M -unsaturated vertices. We have $|L| = |M| + (n - 2|M|) = n - |M|$. By Gallai's Theorem (Theorem 3.1.22), $|M| = \alpha'(G)$ if and only if $|L| = \beta'(G)$.

b) *A minimal edge cover L is a minimum edge cover if and only if it contains a maximum matching.* As observed in proving Theorem 3.1.22, every minimal edge cover consists of disjoint stars. The largest matching contained in a disjoint union of stars consists of one edge from each component. The size of this matching is $n - |L|$. Hence a minimal edge cover L has size $n - \alpha'(G)$ if and only if L contains a matching of size $\alpha'(G)$.

3.1.44. *If G is a simple graph in which the sum of the degrees of any k vertices is less than $n - k$, then every maximal independent set in G has more than k vertices.* Let S be an independent set. If $|S| \leq k$, then the sum of the degrees of the vertices in S is less than $n - k$. This means that some vertex x outside S is not a neighbor of any vertex in S , and hence x can be added to form an independent set containing S . Thus maximal independent sets must have more than k vertices.

3.1.45. *If xy and xz are α -critical edges in G and $y \leftrightarrow z$, then G contains an induced odd cycle (through xy and xz).* Let Y, Z be maximum stable sets in $G - xy$ and $G - xz$, respectively. Since Y, Z are not independent in G , we have $x, y \in Y$ and $x, z \in Z$.

Proof 1. Let $H = G[Y \Delta Z]$. Since $x \in Y \cap Z$, H is a bipartite graph with bipartition $Y - Z, Z - Y$. If some component of H has partite sets of different sizes, then we can substitute the larger for the smaller in Y or Z to obtain a stable set in G of size exceeding $\alpha(G)$.

If y and z belong to different components H_y and H_z of H , then let S be the union of $V(H_y) \cap Z, V(H_z) \cap Y$, one partite set of each other component of H , and $Y \cap Z$. Since $x \in Y \cap Z$ and $y, z \notin S$, the set S is independent in G . Also, $|S| = |Y| = |Z| > \alpha(G)$. Hence y and z belong to the same component of H . A shortest y, z -path in H is a chordless path of odd length in G , and it completes a chordless odd cycle with zx and xy .

Proof 2. Let $H' = G[(Y \Delta Z) \cup \{x\}]$. Note that $|Y \Delta Z|$ is even, since $|Y| = |Z|$. Let $2k = |Y \Delta Z|$. If H' is bipartite, then it has an independent set of size at least $k+1$, which combines with $Y \cap Z - \{x\}$ to form an independent set of size $\alpha(G) + 1$ in G . Hence H' has an odd cycle. Since H is bipartite, this cycle passes through x . Since $N_{H'}(x) = \{y, z\}$, the odd cycle contains the desired edges.

3.1.46. A graph has domination number 1 if and only if some vertex neighbors all others. This is immediate from the definition of dominating set.

3.1.47. The smallest tree where the vertex cover number exceeds the domination number is P_6 . In a graph with no isolated vertices, every vertex cover is a dominating set, since every vertex is incident to an edge, and at least one endpoint of that edge is in the set. Hence $\gamma(G) \leq \beta(G)$. We want a tree where the inequality is strict.

If $\gamma(G) = 1$, then a single vertex is adjacent to all others, and since G is a tree there are no other edges, so $\beta(G) = 1$. Hence we need $\gamma(G) \geq 2$ and $\beta(G) \geq 3$. A tree is bipartite, so $\alpha'(G) = \beta(G) \geq 3$. A matching of size 3 requires at least 6 vertices. There are two isomorphism classes of 6-vertex trees with perfect matchings, and P_6 is the only one having a dominating set of size 2. (Smaller trees can also be excluded by case analysis instead of using the König–Egerváry Theorem.)



3.1.48. $\gamma(C_n) = \gamma(P_n) = \lceil n/3 \rceil$. With maximum degree 2, vertices can dominate only two vertices besides themselves. Therefore, $\lceil n/3 \rceil$ is a lower bound. Picking every third vertex starting with the second (and using the last when n is not divisible by 3) yields a dominating set of size $\lceil n/3 \rceil$.

3.1.49. In a graph G without isolated vertices, the complement of a minimal dominating set is a dominating set, and hence $\gamma(G) \leq n(G)/2$. Let S be a minimal dominating set. Every vertex of $\bar{S}$ has a neighbor in S , and by minimality this fails when a vertex is omitted from S . For each $x \in S$, there is thus a vertex of $\bar{S}$ whose only neighbor in S is x . In particular, every $x \in S$ has a neighbor in $\bar{S}$, which means that $\bar{S}$ is a dominating set.

Since S and $\bar{S}$ are disjoint dominating sets, one of them has size at most $n(G)/2$.

3.1.50. If G is a n -vertex graph without isolated vertices, then $\gamma(G) \leq n - \beta'(G) \leq n/2$.

Proof 1. Since every edge covers at most two vertices, always $\beta'(G) \geq n/2$. As discussed in the proof of Theorem 3.1.22, the components of a minimum edge cover are stars, and the number of stars is $n - \beta'(G)$. Since the union of these stars is a spanning subgraph, choosing the centers of these stars yields a dominating set.

Proof 2. Since $\alpha'(G) = n - \beta'(G)$ by Theorem 3.1.22, it suffices to show that $\gamma(G) \leq \alpha'(G)$. In a maximum matching M , the two endpoints of an edge in M cannot have distinct unsaturated neighbors. Also the unsaturated neighbors all have saturated neighbors. Therefore, picking from

each edge of M the endpoint having unsaturated neighbor(s) (or either if neither has such a neighbor) yields a dominating set of size $\alpha'(G)$.

Construction of n -vertex graphs with domination number k , for $1 \leq k \leq n/2$. Form G from a matching of size k by selecting one vertex from each edge and adding edges to make these vertices pairwise adjacent.

3.1.51. Domination in an n -vertex simple graph G with no isolated vertices.

a) $\lceil n/(1 + \Delta(G)) \rceil \leq \gamma(G) \leq n - \Delta(G)$. Each vertex takes care of itself and at most $\Delta(G)$ others; thus $\gamma(G)(1 + \Delta(G)) \geq n$. For the upper bound, note that the set consisting of all vertices *except* the neighbors of a vertex of maximum degree is a dominating set.

b) $(1 + \text{diam } G)/3 \leq \gamma(G) \leq n - \lceil 2\text{diam } G/3 \rceil$. Let P be a shortest u, v -path, where $d(u, v) = \text{diam } G$. Taking the vertices at distances 1, 4, 7, ... along P from u yields a set that dominates all the vertices of P (v is also needed if $\text{diam } G$ is divisible by 3. Even if all the vertices off P are needed to augment this to a dominating set, we still have used at most $n - \lceil 2\text{diam } G/3 \rceil$ vertices.

For the lower bound, the vertices at distances 0, 3, 6, ... from u along P must be dominated by distinct vertices in a dominating set; a vertex dominating two of them would yield a shorter u, v -path. This yields the lower bound.

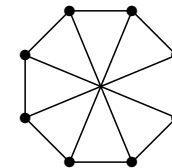
Both bounds hold with equality for the path P_n .

3.1.52. If the diameter of G is at least 3, then $\gamma(\bar{G}) \leq 2$. Let u and v be two vertices such that $d_G(u, v) = 3$. The set $\{u, v\}$ is a dominating set in $\bar{G}$, because u and v have no common neighbors in G . For $x \in V(G) - \{u, v\}$, at least one of $\{u, v\}$ is nonadjacent to x in G and therefore adjacent to it in $\bar{G}$.

3.1.53. Examples with specified domination number.

A $5k$ -vertex graph with domination number $2k$ and minimum degree 2. Begin with kC_5 and add edges to form a cycle using one vertex from each 5-cycle. Two vertices must be used from each original 5-cycle, and this suffices for a dominating set.

A 3-regular graph G with $\gamma(G) = 3n(G)/8$. In a 3-regular graph, each vertex dominates itself and three others. In the 8-vertex graph below, deletion of any vertex and its neighbors leaves P_4 , which cannot be dominated by one additional vertex. Hence the domination number is 3.



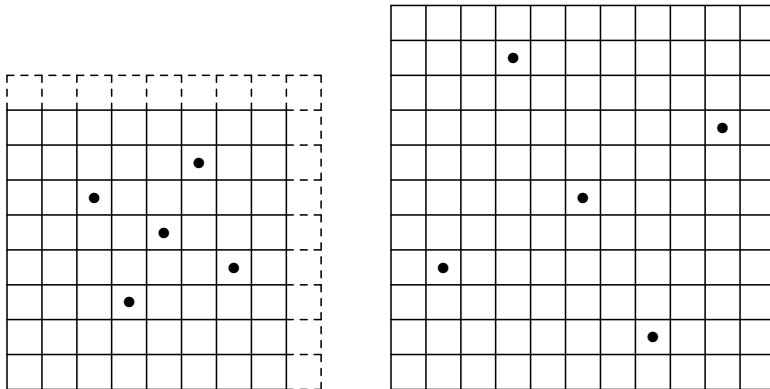
3.1.54. The Petersen graph has domination number 3 and total domination number 4. Each vertex dominates itself and three others, so at least three vertices are needed. Since the graph has diameter 2, the neighbors of a single vertex form a dominating set.

A total dominating set S must include a neighbor of every vertex in S . Hence S must contain two adjacent vertices. This pair leaves four undominated vertices. Adding a neighbor of the original pair dominates at most three of these, since the graph is 3-regular. Hence $|S| \geq 4$. One vertex and its neighbors form a total dominating set of size four.

3.1.55. Dominating sets in the hypercube Q_4 . Since Q_4 is 4-regular, each vertex dominates itself and four others. Now $n(Q_4) = 16$ yields $\gamma(Q_4) \geq \lceil 16/5 \rceil = 4$. Since $\{0000, 0111, 1100, 1011\}$ is an independent dominating set and $\{0000, 0001, 1110, 1111\}$ is a total dominating set, the domination, independent domination, and total domination numbers all equal 4.

Adding two vertices to this total dominating set of size 4 completes a connected dominating set of size 6. We show there is no smaller connected dominating set. A connected 5-vertex subgraph contains two incident edges. Let S be the set of three vertices in two such edges. The set T of vertices undominated by S has size 6. Each neighbor of a vertex of S dominates at most two vertices in T . Each vertex of T dominates at most three vertices in T , except for one vertex that dominates itself and four others (For example, if $S = \{0000, 0001, 0010\}$, then the high-degree vertex of $Q_4[T]$ is 1111.) To dominate T with only two additional vertices, we must therefore use the high-degree vertex of T . However, its distance to S is 3, so it cannot be used to complete a connected set of size 5.

3.1.56. Five pairwise non-attacking queens can control an 8-by-8 chessboard. As shown below, they can also control a 9-by-9 chessboard. Five queens still suffice for an 11-by-11 chessboard, but this configuration does not exist on the 8-by-8 board.



3.1.57. An n -vertex tree with domination number 2 in which the minimum size of an independent dominating set is $\lfloor n/2 \rfloor$. Consider the tree of diameter 3 with two central vertices u and v in which one central vertex has $\lfloor (n-2)/2 \rfloor$ leaf neighbors and the other has $\lceil (n-2)/2 \rceil$ leaf neighbors. The set $\{u, v\}$ is a dominating set, but these cannot both appear in an independent dominating set. If u does not appear in a dominating set, then all its leaf neighbors must appear. We also must include at least one vertex from the set consisting of v and its leaf neighbors, since these are not dominated by the other leaves. Hence the independent dominating set must have at least $\lfloor (n-2)/2 \rfloor + 1$ vertices.

3.1.58. Every $K_{1,r}$ -free graph G has an independent dominating set of size at most $(r-2)\gamma(G) - (r-3)$. Let S be a minimum dominating set in G . Let S' be a maximal independent subset of S . Let $T = V(G) - R$, where R is the set $N(S') \cup S'$ of vertices dominated by S' . Let T' be a maximal independent subset of T .

Since T' contains no neighbor of S' , $S' \cup T'$ is independent. Since S' is a maximal independent subset of S , every vertex of $S - S'$ has a neighbor in S' . Similarly, T' dominates $T - T'$. Hence $S' \cup T'$ is a dominating set.

It remains to show that $|S' \cup T'| \leq (r-1)\gamma(G) - (r-3)$. Each vertex of $S - S'$ has at most $r-2$ neighbors in T' , since it has a neighbor in S' , and $S' \cup T'$ is independent, and G is $K_{1,r}$ -free. Since S is dominating, each vertex of T' has at least one neighbor in $S - S'$. Hence $|T'| \leq (r-2)|S - S'|$, which yields $|S' \cup T'| \leq (r-2)|S| - (r-3)|S'|$. Since $|S| = \gamma(G)$ and $|S'| \geq 1$, we conclude that $|S' \cup T'| \leq (r-2)\gamma(G) - (r-3)$.

3.1.59. In a graph G of order n , the minimum size of a connected dominating set is n minus the maximum number ℓ of leaves in a spanning tree. For the upper bound, deleting the leaves in a spanning tree with ℓ leaves yields a connected dominating set of size $n - \ell$.

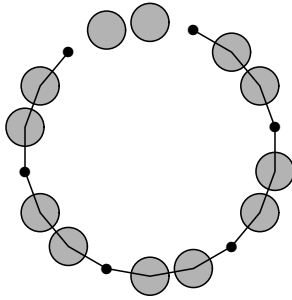
For the lower bound, we form a spanning tree T of G by taking a spanning tree in the subgraph induced by a connected dominating set S and adding each remaining vertex as a neighbor of one of these. Thus $|S|$ is n minus the number of leaves in T . Since this number of leaves is at most ℓ , we have $|S| \leq n - \ell$.

3.1.60. A graph with minimum degree k and no connected dominating set of size less than $3n(G)/(k+1) - 2$. Form G from a cyclic arrangement of $3m$ pairwise-disjoint cliques of sizes $\lceil k/2 \rceil, \lfloor k/2 \rfloor, 1, \lceil k/2 \rceil, \lfloor k/2 \rfloor, 1, \dots$ in order by making each vertex adjacent to every vertex in the clique before it and the clique after it. Each vertex is adjacent to all but itself in its own clique and the two neighboring cliques, so G is k -regular.

A graph has a connected dominating set of size r if and only if it has a spanning tree with at most r non-leaf vertices (Exercise 3.1.59). Hence we show that spanning trees in G must have many non-leaves. Let S be the set of vertices in the cliques of size 1 in the construction.

In a subgraph of G having all pairs of nearest vertices in S joined by paths through the two intervening cliques, there is a cycle. If at least two such pairs are not joined by such paths, then the subgraph is disconnected. Therefore, every spanning tree of G contains a path P directly connecting all but one of the successive pairs, as shown below.

At least one endpoint of P must be a non-leaf. If one endpoint of P is a leaf, then the other endpoint of P and some vertex in the untouched clique next to it must be non-leaves. In either case, we have obtained at least $3m - 2$ non-leaves. Since $m = n/(k + 1)$, we have the desired bound.



3.2. ALGORITHMS AND APPLICATIONS

3.2.1. A weighted graph with four vertices where the maximum weight matching is not a maximum size matching. Let $G = P_4$, and give the middle edge greater weight than the sum of the other weights.

3.2.2. Use of the Hungarian Algorithm to test for the existence of a perfect matching in a bipartite graph G . Given that the partite sets of G have size n , form a weighted matching problem on $K_{n,n}$ in which the edges of G have weight 1 and the edges not in G have weight 0. There is a perfect matching in G if and only if the solution to the weighted matching problem is n .

3.2.3. Multiplicity of stable matchings. With men u, v and women a, b , there may be two stable matching. Suppose the preferences are $u : a > b$, $v : b > a$, $a : v > u$, $b : u > v$. If both men get their first choices, then they prefer no one to their assigned partner, so the matching is stable. The same argument applies when the women get their first choices. However,

the matchings with men getting their first choices and women getting their first choices are different.

3.2.4. Stable matchings under proposal algorithm. Consider the preference orders listed below.

Men $\{u, v, w, x, y, z\}$	Women $\{a, b, c, d, e, f\}$
$u : a > b > d > c > f > e$	$a : z > x > y > u > v > w$
$v : a > b > c > f > e > d$	$b : y > z > w > x > v > u$
$w : c > b > d > a > f > e$	$c : v > x > w > y > u > z$
$x : c > a > d > b > e > f$	$d : w > y > u > x > z > v$
$y : c > d > a > b > f > e$	$e : u > v > x > w > y > z$
$z : d > e > f > c > b > a$	$f : u > w > x > v > z > y$

When men propose, the steps of the algorithm are as below. For each round, we list the proposals by u, v, w, x, y, z in order, followed by the resulting rejections. Round 1: a, a, c, c, c, d ; $a \times v$, $c \times w$, $c \times y$. Round 2: a, b, b, c, d, d ; $b \times v$, $d \times z$. Round 3: a, c, b, c, d, e ; $c \times x$. Round 4: a, c, b, a, d, e ; $a \times u$. Round 5: b, c, b, a, d, e ; $b \times u$. Round 6: d, c, b, a, d, e ; $d \times u$. Round 7: c, c, b, a, d, e ; $c \times u$. Round 8: f, c, b, a, d, e ; stable matching.

When women propose, the steps of the algorithm are as below. For each round, we list the proposals by a, b, c, d, e, f in order, followed by the resulting rejections. Round 1: z, y, v, w, u, u ; $u \times e$. Round 2: z, y, v, w, v, u ; $v \times e$. Round 3: z, y, v, w, x, u ; stable matching.

Note that the pairs uf and vc occur in both results, and in all other cases the women are happier when the women propose and the men are happier when the men propose.

3.2.5. Maximum weight transversal. For each matrix below, we underscore a maximum weight transversal, and the labels on the rows and columns form a cover whose total cost equals the weight of the transversal.

For every position (i, j) , the label on row i plus the label on column j is at least the entry in position (i, j) in the matrix. Hence the labeling is feasible for the dual problem. Equality between the sum of the labels in a feasible labeling and the sum of the entries of a transversal implies that the transversal is one of maximum weight and feasible labeling is one of minimum weight, because every feasible labeling has sum as large as the weight of every matching (since the positions in the matching must be covered disjointly by the labels).

	0	0	1	0	2		3	1	2	1	0		0	0	1	1	2
4	<u>4</u>	4	4	3	6	7	<u>7</u>	8	<u>9</u>	8	7	3	<u>1</u>	2	3	4	<u>5</u>
3	1	1	<u>4</u>	3	4	6	8	7	<u>6</u>	<u>7</u>	6	7	<u>6</u>	<u>7</u>	8	7	2
4	1	<u>4</u>	5	3	5	6	9	6	5	4	<u>6</u>	3	1	3	<u>4</u>	4	5
7	5	6	4	7	<u>9</u>	5	<u>8</u>	5	7	6	4	7	<u>3</u>	6	2	<u>8</u>	7
8	5	3	6	<u>8</u>	3	5	<u>7</u>	<u>6</u>	5	5	5	4	<u>4</u>	1	3	5	4

Review of the “Hungarian Algorithm” for maximum weighted matching in the assignment problem. Find a feasible vertex labeling for the dual, i.e. weights $l(v)$ such that $l(x_i) + l(y_j) \geq w(ij)$. (This can be done by using the maximum in each row as the row label, with 0's for the columns.) Subtract out to find the “excess value” matrix $l(x_i) + l(y_j) - w(ij)$. Find a maximum matching and minimum cover in the equality subgraph (0's in the excess matrix). If this is a perfect matching, its value equals the dual value $\sum l(v)$ being minimized, hence is optimal. If not, let S be the set of rows not in the cover, T the set of columns in the cover, and ε the minimum excess value in the uncovered positions. Subtract ε from the row labels in S , add ε to the row labels in T , readjust the excess matrix, and iterate. (Note: any minimum cover can be used, and we know from the proof of the König–Egerváry Theorem that we can obtain a minimum cover by using $T \cup (X - S)$, where T and S are the subsets of Y and X reachable by alternating paths from the unsaturated vertices in the row-set X .)

If the matching was not complete, then $|S| > |T|$ and $\sum l(v)$ decreases, which guarantees the finiteness of the algorithm. The positions are of four types, corresponding to edges from S to T , $X - S$ to T , S to $Y - T$, and $X - S$ to $Y - T$. The change to the excess in the four cases is 0, $+\varepsilon$, $-\varepsilon$, 0, respectively. Note that ε was defined to be the minimum excess corresponding to edges from S to $Y - T$, so every excess remains positive. For the first matrix above, the successive excess matrices computed in the algorithm could look like those below. These are not unique, because different matchings could be chosen in the equality subgraphs. The entries in the matching and the rows and columns in the cover ($X - S$ and T) are indicated with underscores.

$$\begin{array}{c} \begin{array}{ccccc} 0 & 0 & \underline{0} & 0 & \underline{0} \\ 6 & \begin{pmatrix} 2 & 2 & 2 & 3 & 0 \end{pmatrix} \\ 4 & \begin{pmatrix} 3 & 3 & \underline{0} & 1 & 0 \end{pmatrix} \\ 5 & \begin{pmatrix} 4 & 1 & 0 & 2 & \underline{0} \end{pmatrix} \\ 9 & \begin{pmatrix} 4 & 3 & 5 & 2 & 0 \end{pmatrix} \\ \underline{8} & \begin{pmatrix} 3 & 5 & 2 & \underline{0} & 5 \end{pmatrix} \end{array} & \rightarrow & \begin{array}{ccccc} 0 & 0 & 1 & 0 & \underline{1} \\ 5 & \begin{pmatrix} 1 & 1 & 2 & 2 & 0 \end{pmatrix} \\ \underline{3} & \begin{pmatrix} 2 & 2 & \underline{0} & 0 & 0 \end{pmatrix} \\ 4 & \begin{pmatrix} 3 & \underline{0} & 0 & 1 & 0 \end{pmatrix} \\ 8 & \begin{pmatrix} 3 & 2 & 5 & 1 & \underline{0} \end{pmatrix} \\ \underline{8} & \begin{pmatrix} 3 & 5 & 3 & \underline{0} & 6 \end{pmatrix} \end{array} & \rightarrow & \begin{array}{ccccc} 0 & 0 & 1 & 0 & 2 \\ 4 & \begin{pmatrix} \underline{0} & 0 & 1 & 1 & 0 \end{pmatrix} \\ 3 & \begin{pmatrix} 2 & 2 & \underline{0} & 0 & 1 \end{pmatrix} \\ 4 & \begin{pmatrix} 3 & \underline{0} & 0 & 1 & 1 \end{pmatrix} \\ 7 & \begin{pmatrix} 2 & 1 & 4 & 0 & \underline{0} \end{pmatrix} \\ 8 & \begin{pmatrix} 3 & 5 & 3 & \underline{0} & 7 \end{pmatrix} \end{array} \end{array}$$

3.2.6. Finding a transversal of minimum weight. Let the rows correspond to vertices $x_1, \dots, x_5$, the columns to vertices $y_1, \dots, y_5$, and let the weight of edge $x_i y_j$ be the value in position ij . Optimality of the answer can be proved by exhibiting an optimal matching and exhibiting a feasible labeling for the dual problem that has the same total value.

Alternatively, finding a minimum transversal is the same as finding a minimum weight perfect matching in the corresponding graph, which corresponds to a maximum weight matching in the weighted graph obtained by subtracting all the weights from a fixed constant. In the example given,

we could subtract the weights from 13, and then the answer would be $5 \cdot 13$ minus the maximum weight of a transversal in the resulting matrix.

In the direct approach, the dual problem is to maximize $\sum l(v)$ subject to $l(x_i) + l(y_j) \leq w(x_i y_j)$. Subtracting the labels from the weights yields the “reduced cost” matrix. At each iteration, we determine the equality subgraph and ε as before, but this time add ε to the labels of vertices in S (rows not in the cover) and subtract ε from the labels of vertices in T (columns in the cover). Since $|S| > |T|$, $\sum l(v)$ increases. Every matching has weight at least $\sum l(v)$. When G_l contains a complete matching, $\min \sum w$ and $\max \sum l$ are attained and equal.

In the matrix below, the underscored positions form a minimum-weight transversal; the weight is 30. In the dual problem, the indicated labeling has total value 30, and the labels $l(x_i)$ and $l(y_j)$ sum to at most the matrix entry $w_{i,j}$. Hence these solutions are optimal.

$$\begin{array}{ccccc} & 0 & 1 & 3 & 5 & 2 \\ \begin{array}{c} 4 \\ 2 \\ 4 \\ 5 \\ 4 \end{array} & \begin{pmatrix} \underline{4} & 5 & 8 & 10 & 11 \\ 7 & 6 & \underline{5} & 7 & 4 \\ 8 & \underline{5} & 12 & 9 & 6 \\ 6 & 6 & 13 & 10 & \underline{7} \\ 4 & 5 & 7 & \underline{9} & 8 \end{pmatrix} \end{array}$$

3.2.7. The Bus Driver Problem. Bus drivers are paid overtime for the time by which their routes in a day exceed t . There are n bus drivers, n morning routes with durations $x_1, \dots, x_n$, and n afternoon routes with durations $y_1, \dots, y_n$. Assign to the edge $a_i b_j$ the weight $w_{i,j} = \max\{0, x_i + y_j - t\}$. The problem is then to find the perfect matching of minimum total weight. Index the morning runs so that $x_1 \geq \dots \geq x_n$. Index the afternoon runs so that $y_1 \geq \dots \geq y_n$. A feasible solution matches a_i to $b_{\sigma(i)}$ for some permutation σ of $[n]$. If there exists $i < j$ with $\sigma(i) > \sigma(j)$, then we have

$$\alpha = w_{i,\sigma(i)} + w_{j,\sigma(j)} = \max\{0, x_i + y_{\sigma(i)} - t\} + \max\{0, x_j + y_{\sigma(j)} - t\}$$

$$\beta = w_{i,\sigma(j)} + w_{j,\sigma(i)} = \max\{0, x_i + y_{\sigma(j)} - t\} + \max\{0, x_j + y_{\sigma(i)} - t\}$$

It suffices to prove that $\alpha \geq \beta$, because then there exists a minimizing permutation with no inversion. The nonzero terms in the maximizations have the same sum for each pair. Also,

$$x_i + y_{\sigma(i)} - t \geq x_i + y_{\sigma(j)} - t \geq x_j + y_{\sigma(j)} - t$$

$$x_i + y_{\sigma(i)} - t \geq x_j + y_{\sigma(i)} - t \geq x_j + y_{\sigma(j)} - t$$

If the central terms in the inequalities are both positive, then α is at least their sum, which equals β . If both are nonpositive, then $\alpha \geq 0 = \beta$. If the first is positive and the second nonpositive, then

$$\alpha = x_i + y_{\sigma(i)} - t \geq x_i + y_{\sigma(j)} - t = \beta.$$

If instead the second is positive, then

$$\alpha = x_i + y_{\sigma(i)} - t \geq x_j + y_{\sigma(i)} - t = \beta.$$

3.2.8. When the weights in a matrix are the products of nonnegative numbers associated with the rows and columns, a maximum weight transversal is obtained by pairing the row having the k th largest row weight with the column having the k th largest column weight, for each k . We show that all other pairings are nonoptimal. If the weights are not matched in order, then there exist indices i, j such that $a_i > a_j$ but the weight b matched with a_i is less than the weight b' matched with a_j . To show that switching these assignments increases the total weight, we compute

$$\begin{aligned} a_i b' + a_j b &= a_i b + a_i (b' - b) + a_j (b - b') + a_j b' \\ &= a_i b + a_j b' + (a_i - a_j)(b' - b) > a_i b + a_j b' \end{aligned}$$

When the weights in a matrix are the sums of nonnegative numbers associated with the rows and columns, every transversal has the same weight. Since a transversal uses one element in each row and each column, $w_{i,j} = a_i + b_j$ means that every transversal has total weight $\sum a_i + \sum b_j$.

3.2.9. *One-sided preferences.* There are k seminars and n students, each student to take one seminar. The i th seminar will have k_i students, where $\sum k_i = n$. Each student ranks the seminars; we seek a stable assignment where no two students can both improve by switching.

Form an X, Y -bigraph where X is the set of students and Y has k_i vertices for each vertex i . For each edge from student x_j to a vertex representing the i th seminar, let the weight be k minus the rank of the i th seminar in the preference of x_j .

A maximum weight matching in this weighting is a stable assignment, since if two students can both improve by trading assignments, then the result would be a matching of larger weight.

3.2.10. *Weighted preferences need not be stable.* Consider men $\{x_1, x_2, x_3\}$ and women $\{y_1, y_2, y_3\}$. Each assigns $3 - i$ points to the i th person in his or her preference list. Hence we indicate a preference order by a triple whose entries are 0, 1, 2 in some order, with the position of the integer i being the index of the person of the opposite sex to whom this person assigns i points.

In the matrix below, the preference vectors of the men and women label the rows and columns, respectively. An entry in the matrix is the sum

of the points assigned to that potential edge by the two people. The underlined diagonal is the only matching that uses the maximum entry in each row and column, so it is the only maximum-weight matching. However, it is not a stable matching, because man x_1 and woman y_1 prefer each other to their assigned mates.

$$\begin{array}{ccc} & 120 & 120 & 210 \\ \begin{array}{l} 120 \\ 021 \\ 120 \end{array} & \left(\begin{array}{ccc} \underline{2} & 3 & 3 \\ 1 & 4 & 2 \\ 2 & 2 & 0 \end{array} \right) \end{array}$$

The example can be extended for all larger numbers of men and women by adding pairs who are each other's first choice and are rated last by the six people in this example.

3.2.11. *In the result of the Gale-Shapley Proposal Algorithm with men proposing, every man receives a mate at least as high on his list as in any other stable matching.* We prove that under the G-S Algorithm with men proposing, no man is ever rejected by any woman who is matched to him in any stable matching. This yields the result, since each man's sequence of proposals proceeds downward from the top of his list, and he can only wind up with a woman less desirable than his most desirable match over all stable matchings if he is rejected by some women who matches him in some stable matching.

Consider the first time when some man x is rejected by a woman a to whom he is matched in some stable matching M . The rejection occurs because a has a proposal from a man y higher than x on her list. In M , man y is matched to some woman b . Since M is stable, y cannot prefer a to b . Thus b appears above a on the list for y . But now the decreasing property of proposals from men implies that y has proposed to b in the G-S Algorithm before proposing to a . If y is now proposing to a , then y was previously rejected by b . Since y is matched to b in M , this contradicts our hypothesis that a rejecting x was the first rejection involving a pair that occurs in some stable matching.

3.2.12. *The Stable Roommates Problem defined by the preference orderings below has no stable matching.* There are only three matchings to consider: $ab|cd$, $ac|bd$, and $ad|bc$. In each, two non-paired people prefer each other to their current roommates. The problematic pairs are $\{b, c\}$, $\{a, b\}$, $\{a, c\}$ in the three matchings, respectively.

$$\begin{aligned} a &: b > c > d \\ b &: c > a > d \\ c &: a > b > d \\ d &: a > b > c \end{aligned}$$

3.2.13. In the stable roommates problem, suppose that each individual declares a top portion of the preference list as “acceptable”. Define the *acceptability graph* to be the graph whose vertices are the people and whose edges are the pairs of people who rank each other as acceptable. Prove that all sets of rankings with acceptability graph G lead to a stable matching if and only if G is bipartite. (Abeledo–Isaak [1991]).

In the stable roommates problem with each individual declaring a top portion of the preference list as “acceptable”, and the acceptability graph being the graph on the people whose edges are the mutually acceptable pairs, all sets of rankings with acceptability graph G allow stable matchings if and only if G is bipartite. If G is bipartite, then we view the two partite sets as the two groups in the classical stable matching problem (isolated vertices may be added to make the partite sets have equal size). The unacceptable choices for an individual x may be put in any order, since they are all (equally) unacceptable, so we can ensure that all choices for x that are in the same partite set appear at the bottom of the preference order for x . In the outcome of the Gale-Shapley Proposal Algorithm, there is no pair (x, a) from opposite partite sets such that a and x prefer each other to their assigned mates. Also no x prefers an individual in its own partite set to the person assigned to x , since all individuals in its own partite set are unacceptable. Hence the stable matching produced for the bipartite version is also stable in the original problem.

If G is not bipartite, then G has an odd cycle $[x_1, \dots, x_k]$. Define a set of rankings such that x_i prefers x_{i+1} to x_{i-1} (indices modulo k), and x_i prefers x_{i-1} to all others. The preferences of people not on the cycle are irrelevant. Since the cycle has odd length, the people on the cycle cannot be paired up using edges of the cycle. Given a candidate matching M , we may assume by symmetry that x_1 is not matched to x_2 or to x_k in M . Now x_1 prefers x_2 to M -mate of x_1 , and x_2 prefers x_1 to the M -mate of x_2 (which might be x_3). Hence the matching M is not stable. Thus there is no stable matching for these preferences, which means that this acceptability graph does not always permit a stable matching.

3.2.14. In the Proposal Algorithm with men proposing, no man is ever rejected by all the women.

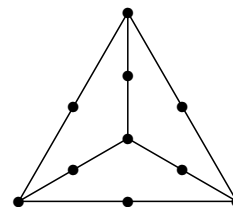
Proof 1. By Theorem 3.2.18, the Proposal Algorithm succeeds, so it ends with each man being accepted before being rejected by all women.

Proof 2. Once a woman has received a proposal, she thereafter receives a proposal on each round, since the key observation is her sequence of “maybe”s is nondecreasing in her list. If a round has j rejections and $n - j$ “maybe”s, then the $n - j$ unrejected men are distinct, since men propose to exactly one woman on each round.

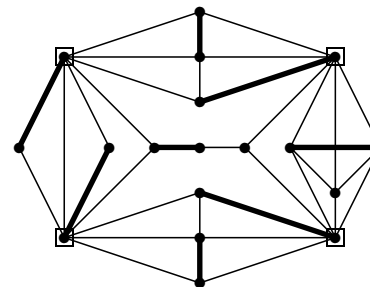
When a man has been rejected by k women, those k women have received proposals, and thereafter by the remarks above they always receive proposals from k distinct men. In particular, when a man has been rejected by $n - 1$ women, on the next round they receive proposals from $n - 1$ distinct men other than him, and he proposes to the remaining women, so the algorithm ends successfully on that step.

3.3. MATCHINGS IN GENERAL GRAPHS

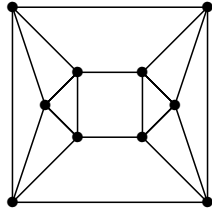
3.3.1. The graph G below has no 1-factor. Deleting the four vertices with degree 3 leaves six isolated vertices; thus $o(G - S) > |S|$ for this set S .



3.3.2. The maximum size of a matching in the graph G below is 8. A matching of size 8 is shown. Since $n(G) = 18$, it suffices to show that G has no perfect matching. For this we present a set S such that $o(G - S) > |S|$, violating Tutte’s condition. Such a set S is marked. (Note: the smallest vertex cover has size 9, so duality using vertex cover is not adequate.)



3.3.3. k -factors in the 4-regular graph below. The full graph is a 4-factor, and the spanning subgraph with no edges is a 0-factor. There is a 2-factor consisting of the outer 4-cycle and the 6-cycle on the remaining vertices. Since these cycles have even length, taking alternating edges from both cycles yields a 1-factor. Deleting the edges of the 1-factor leaves a 3-factor.



3.3.4. A k -regular bipartite graph is r -factorable if and only if r divides k . The edges incident to a single vertex demonstrate necessity. For sufficiency, a k -regular bipartite graph has a perfect matching, and hence by induction on k is 1-factorable; take unions of the 1-factors in groups of r .

3.3.5. Join of graphs G and H . As long as G and H have at least one vertex each, $G \vee H$ is connected (it has $K_{n(G), n(H)}$ as a spanning subgraph).

In forming $G \vee H$, every vertex of G gains $n(H)$ neighbors in H , and every vertex in H gains $n(G)$ neighbors in G . Hence $\Delta(G \vee H) = \max\{\Delta(G) + n(H), \Delta(H) + n(G)\}$.

3.3.6. A tree T has a perfect matching if and only if $o(T - v) = 1$ for every $v \in V(T)$. *Necessity.* Let M be a perfect matching in T in which u is the vertex matched to v . Each component of $T - v$ not containing u must have a perfect matching and hence even order. The component containing u is matched by M except for u , so it has odd order.

Sufficiency. **Proof 1** (construction of matching). Suppose that $o(T - v) = 1$ for all $v \in V(T)$. Each vertex has a neighbor in one component of odd order. We claim that pairing each w to its neighbor in the odd component of $T - w$ yields a matching. It suffices to prove that if u is the neighbor of v in the unique odd component T_1 of $T - v$, then v is the neighbor of u in the unique odd component T_2 of $T - u$. Since $o(T - v) = 1$, the components of $T - v$ other than T_1 have even order. The subtree T_2 consists of these components and edges from these to v . Hence T_2 includes some even vertex sets and $\{v\}$, and T_2 thus has odd order.

Proof 2 (induction on $n(T)$). The claim is immediate for $n(T) = 2$. If $n(T) > 2$ and $o(T - v) = 1$ for all v , then the neighbor w of any leaf u has only one leaf neighbor. Let $T' = T - \{u, w\}$. The components of $T' - v$ are the same as the components of $T - v$, except that one of them in $T - v$ includes $\{u, w\}$ and the corresponding component of $T' - v$ omits them. Hence the parities are the same, and $o(T' - v) = 1$ for all $v \in V(T')$. By the induction hypothesis, T' has a perfect matching, and adding the edge uw to this completes a perfect matching in T .

(Comment: It is also possible to do the induction step by deleting an arbitrary vertex, but it is then a bit more involved to prove that every

component T' of the forest left by matching v to its neighbor in the odd component of $T - v$ satisfies the condition $o(T' - x) = 1$ for all x .

Proof 2a (induction and extremality). The basis again is $n(T) = 2$. For $n(T) > 2$, let P be a longest path. Let x be an endpoint of P , with neighbor y . Since $o(T - y) = 1$ and P is a longest path, $d_T(y) = 2$. Deleting x and y yields a tree T' such that $o(T' - v) = o(T - v) = 1$ for all $v \in V(T')$, since x and y lie in the same component of $T' - v$. Hence the induction hypothesis yields a perfect matching in T' , which combines with xy to form a perfect matching in T .

Proof 3 (Tutte's Condition). By Tutte's Theorem, it suffices to prove for all $S \subseteq V(T)$ that $o(T - S) \leq |S|$. We prove this by induction on $|S|$. Since $o(T - v) = 1$, we have $n(T)$ even, and hence $o(T - \emptyset) = 0$. When $|S| = 1$, the hypothesis $o(T - v) = 1$ yields the desired inequality for $S = \{v\}$.

For the induction step, suppose that $|S| > 1$. Let T' be the smallest subtree of T that contains all of S . Note that all leaves of T' are elements of S . Let v be a leaf of T' , and let $S' = S - \{v\}$. By the induction hypothesis, $o(T - S') \leq |S'| = |S| - 1$. It suffices to show that when we delete v from $T - S'$, the number of odd components increases by at most 1.

Let T'' be the component of $T - S'$ containing v . Deleting v from $T - S'$ replaces T'' with the components of $T'' - v$. We worry only if $T'' - v$ has at least two odd components. Since v is a leaf of T' , all of S' lies in one component of $T - v$. Hence the components of $T'' - v$ are the same as the components of $T - v$ except for the one component of $T - v$ containing S' .

Since $o(T - v) = 1$, we can have two odd components in $T'' - v$ only if the one odd component of $T - v$ is a component of $T'' - v$ and the component of $T'' - v$ that is not a component of $T - v$ is also odd. Since the remaining components of $T'' - v$ are even, this means that T'' itself has odd order (it includes v and the two odd components of $T'' - v$). Therefore, the replacement of T'' with the components of $T'' - v$ increases the number of odd components only by one. We conclude that $o(T - S) \leq o(T - S') + 1 \leq |S'| + 1 = |S|$, which completes the induction step.

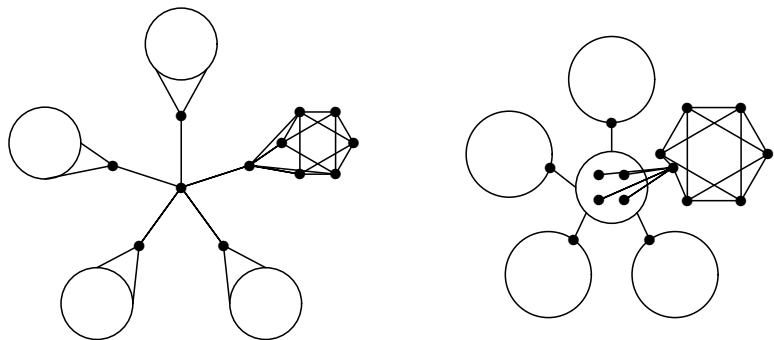
3.3.7. There exist k -regular simple graphs with no perfect matching. When k is even, K_{k+1} is a k -regular graph with no perfect matching, since it has an odd number of vertices. When k is odd, there are two usual types of constructions.

Construction 1. Begin with k disjoint copies of K_{k+1} . Delete $(k-1)/2$ disjoint edges from each copy, which drops the degree of $k-1$ vertices in each copy to $k-1$. Add a new vertex v_i to the i th copy, joining it to each of these vertices of degree $k-1$. Add one final vertex x joined to $v_1, \dots, v_k$. The graph has been constructed to be k -regular. Deleting x leaves k components of order $k+2$ (odd); hence the graph fails Tutte's

condition and has no perfect matching.

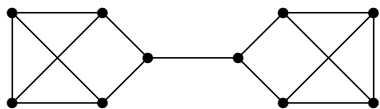
A slight variation is to start with k copies of $K_{k-1,k}$, add a matching of size $(k-1)/2$ to the larger side in each copy, and join the leftover vertices from each larger side to a final vertex x .

Construction 2. Begin with k disjoint copies of K_{k+1} . Subdivide one edge in each copy, which introduces k new vertices of degree 2. To raise their degree to k , add an independent set of $k-2$ additional vertices in the center joined to each of these k vertices. Deleting the $k-2$ vertices in the center violates Tutte's condition.



3.3.8. No graph with a cut-vertex is 1-factorable. Suppose v is a cut-vertex of G . If G is 1-factorable, then G has even order, and $G - v$ has a component H of odd order. For any 1-factor using an edge incident to v whose other endpoint is not in H , the vertices of H cannot all be matched. The contradiction implies there is no 1-factorization.

A 3-regular simple graph having a 1-factor and connectivity 1.



3.3.9. Every graph G with no isolated vertices has a matching of size at least $n(G)/(1 + \Delta(G))$. We use induction on the number of edges. In the induction step, we will delete an edge whose endpoints have degree at least 2 (other edge deletions would isolate a vertex). This tells us what we need to cover in the basis step.

Basis step: every edge of G is incident to a vertex of degree 1. In such a graph, every component has at most one vertex of degree exceeding 1, and thus each component is a star. We form a matching using one edge from each component. Since the number of vertices in each component is 1

plus the degree of the central vertex, the number of components is at least $n(G)/(1 + \Delta(G))$.

Induction step: G has an edge e whose endpoints have degree at least 2. Since $G' = G - e$ has no isolated vertex, we can apply the induction hypothesis to obtain $\alpha'(G) \geq \alpha'(G') \geq n(G')/(1 + \Delta(G')) \geq n(G)/(1 + \Delta(G))$.

3.3.10. The maximum possible value of $\beta(G)$ in terms of $\alpha'(G)$ is $2\alpha'(G)$. If G has a maximal matching of size k , then the $2k$ endpoints of these edges form a set of vertices covering the edges, because any uncovered edge could be added to the matching. Hence $\beta(G) \leq 2\alpha'(G)$. A graph consisting of k disjoint triangles has $\alpha' = k$ and $\beta = 2k$, so the inequality is best possible. These values also hold for the graph K_{2k+1} , since we cannot omit two vertices from a vertex cover of K_{2k+1} . More generally, every disjoint union of cliques of odd order satisfies $\beta(G) = 2\alpha'(G)$.

3.3.11. A graph G has a matching that saturates a set $T \subseteq V(G)$ if and only if for all $S \subseteq V(G)$, the number of odd components of $G - S$ contained in $G[T]$ is at most $|S|$.

Necessity. Saturating T requires saturating each vertex in the odd components of $G[T]$, which uses a vertex of S for each such component.

Sufficiency. Form G' by adding a set U of $n(G)$ new vertices adjacent to each other and to every vertex of $G - T$. We claim that G' satisfies Tutte's Condition. Each $S' \subseteq V(G')$ that contains all of U has size at least $n(G)$. Since $G' - S'$ has at most $n(G)$ vertices, it has at most $|S'|$ odd components.

When $U \not\subseteq S'$, what remains of G' outside of T is a single component. Letting $S = V(G) \cap S'$, the number of odd components in $G' - S'$ is thus at most one more than the number of odd components of $G - S$ contained in T . This yields $o(G' - S') \leq |S'| + 1$, which suffices since $n(G')$ is even.

Since G' satisfies Tutte's Condition, G' has a perfect matching. The edges used to saturate T all lie in G , since no edges were added from T to U . Hence these edges form a matching in G that saturates T .

3.3.12. Extension of König–Egerváry to general graphs. A generalized cover of G is a collection of vertex subsets $S_1, \dots, S_k$ and T such that each S_i has odd size and every edge of G has one endpoint in T or both endpoints in some S_i . The weight of a generalized cover is $|T| + \sum(|S_i| - 1)/2$.

The minimum weight $\beta^*(G)$ of a generalized cover equals the maximum size $\alpha'(G)$ of a matching. Always $\alpha'(G) \leq \beta^*(G)$, because a matching uses at most $(|S_i| - 1)/2$ edges within S_i and at most $|T|$ edges incident to T , and there are no edges not of this type when $S_1, \dots, S_k$ and T form a generalized cover. For equality, it suffices to exhibit a generalized cover with weight equal to $\alpha'(G)$.

The Berge–Tutte formula says that $2\alpha'(G) = \min_T \{n - d(T)\}$, where $d(T) = o(G - T) - |T|$ is the deficiency of a vertex set T . Let T be a maximal

set among those having maximum deficiency. For this choice of T , there are no components of (positive) even order in $G - T$, since we could add to T a leaf of a spanning tree of such a component to obtain a larger set T' with the same deficiency. Let $S_1, \dots, S_k$ be the vertex sets of the components of $G - T$. By construction, this is a generalized cover. Because $2\alpha'(G) = n - d(T)$, we have $k = |T| + d(T)$. Thus

$$\beta^*(G) = |T| + \sum_{i=1}^k (|S_i| - 1)/2 = (n + |T| - k)/2 = \alpha'(G).$$

3.3.13. Proof of Tutte's Theorem from Hall's Theorem. Given a graph G such that $o(G - S) \leq |S|$ for all $S \subseteq V(G)$, we prove that G has a perfect matching. Let T be a maximal vertex subset such that $o(G - T) = |T|$.

a) *Every component of $G - T$ is odd, and $T \neq \emptyset$.* If $G - T$ has an even component C , then let v be a leaf of a spanning tree of C . Now $|T \cup \{v\}| = |T| + 1 = o(G - T) + 1 = o(G - (T \cup \{v\}))$, which contradicts the maximality of T . Thus $G - T$ has no even components.

Since $o(G - \emptyset) \leq 0$, the graph G has no odd components. Since $G - T$ has no even components, we have $|T| > 0$, and $G - T$ is smaller than G .

b) *If C is a component of $G - T$, then Tutte's Condition holds for every subgraph of C obtained by deleting one vertex.* Since $C - x$ has even order, a violation requires $o(C - x - S) \geq |S| + 2$. Adding this inequality to $|T| = o(G - T)$ and $|\{x\}| = 1$ yields

$$|T \cup x \cup S| \leq o(G - T) - 1 + o(C - x - S) = o(G - T - x - S),$$

which contradicts the maximality of T .

c) *The bipartite graph H formed from G by contracting the components of $G - T$ (and deleting edges within T) satisfies Hall's Condition for a matching that saturates the partite set opposite T .* There is an edge from a vertex $t \in T$ to a component C of $G - T$ if and only if $N_G(t)$ contains a vertex of C . For $A \subset \mathbf{C}$, let $B = N_H(A)$. The elements of A are odd components of $G - B$; hence $|A| \leq o(G - B)$. Since Tutte's condition yields $o(G - B) \leq |B|$, we have $|N_H(A)| \geq |A|$.

d) *The final proof.* By Hall's Theorem and part (c), H has a matching that saturates $\mathbf{C}$. This matching yields $o(G - T) = |T|$ pairwise disjoint edges from odd components of $G - T$ to T . By part (a), these are all the components of $G - T$. These edges saturates one vertex from each component of $G - T$. By part (c) and the induction hypothesis, the vertices remaining in each component of $G - T$ are saturated by a perfect matching of that subgraph. The union of the matchings created is a perfect matching of G .

3.3.14. *If G is a simple graph with $\delta(G) \geq k$ and $n(G) \geq 2k$, then $\alpha'(G) \geq k$.* Let $n = n(G)$. By the Berge–Tutte Formula, it suffices to show that the deficiency $o(G - S) - |S|$ is at most $n - 2k$ for every $S \subseteq V(G)$. We prove this by contradiction; suppose that $o(G - S) - |S| > n - 2k$.

Let $s = |S|$. We have $o(G - S) > n - 2k + s$. Thus there are more than $n - 2k + s$ vertices outside S . Together with S , we have $n > n - 2k + 2s$. Thus $s < k$. With $s < k$, a vertex outside S has fewer than k neighbors in S , and $\delta(G) \geq k$ implies that no odd components of $G - S$ are single vertices.

Indeed, every component of $G - S$ has at least $1 + k - s$ vertices. Thus we can improve our earlier inequality: $(1 + k - s)(n - 2k + s + 1) + s \leq n$. This simplifies to $(k - s)(n - 2k + s - 1) < 0$. Since $n \geq 2k$, both factors on the left are positive, which yields a contradiction.

3.3.15. *Every 3-regular graph G with at most two cut-edges has a 1-factor.* Since G has at most two cut-edges, at most two odd components of $G - S$ have one edge to S ; the remainder have at least three edges to S (using the parity of degrees). With $|[S, \bar{S}]| = m$, this yields $3|S| \geq m \geq 3o(G - S) - 4$. Thus $|S| \geq o(G - S) - 4/3$. Since $n(G)$ is even, $|S|$ and $o(G - S)$ have the same parity, which means that $o(G - S)$ exceeds $|S|$ only if it is greater by at least 2. This contradicts $o(G - S) \leq |S| + 4/3$. Hence Tutte's condition holds, and Tutte's Theorem implies that G has a 1-factor.

3.3.16. *If G is k -regular and remains connected when any $k - 2$ edges are deleted, then G has a 1-factor.* By Tutte's Theorem, it suffices to show that $o(G - S) \leq |S|$ for every $S \subseteq V(G)$. This follows for $S = \emptyset$ from the assumption that $n(G)$ is even; hence we may assume that $S \neq \emptyset$. Let H be an odd component of $G - S$, and let m be the number of edges joining H to S . In the subgraph H , the sum of the degrees is $kn(H) - m$. Since this must be even and $n(H)$ is odd, k and m must have the same parity.

By the hypothesis, there are at least $k - 1$ edges between H and S . The requirement of equal parity thus yields $m \geq k$. Summing over all odd components of $G - S$ yields at least $k \cdot o(G - S)$ edges between S and $V(G) - S$. Since the degree sum of the vertices in S is exactly $k|S|$, we obtain $k \cdot o(G - S) \leq k|S|$, or $o(G - S) \leq |S|$.

3.3.17. *Under the conditions of Exercise 3.3.16, each edge belongs to some 1-factor in G .* We want to show that $G' = G - x - y$ has a 1-factor. By Tutte's Theorem, since G has even order, it suffices to show that $o(G' - S') \leq |S'| + 1$ for all $S' \subset V(G')$. Equivalently, $o(G - S) \leq |S| - 1$ for all $S \subseteq V(G)$ that contain $\{x, y\}$.

Let l be the number of edges between S and an odd component H of $G - S$; the hypothesis yields $l \geq k - 1$. The sum $kn(H) - m$ of the vertex degrees in H must be even, but $n(H)$ is odd, so k and m must have the same parity; we conclude that $l \geq k$. Summing over all odd components of $G - S$, we have $m \geq k \cdot o(G - S)$, where m is the number of edges between S and the rest of the graph. Since G is k -regular and $G[S]$ contains the edge xy , we have $m \leq k|S| - 2$. Thus $o(G - S) \leq |S| - 2/k$. Since $o(G - S)$ and $|S|$ are integers, we have the needed inequality.

3.3.18. *Construction of a k -regular graph with no 1-factor (when k is odd), such that deleting any $k - 3$ edges leaves a connected graph.* We make the graph simple and connected under the deletion of any $k - 3$ vertices, which is a stronger requirement.

The Tutte set S will have size $k - 2$, leaving k components in $G - S$. Each component of $G - S$ consists of $K_{k-2, k-1}$ plus a cycle added through the vertices in the larger partite set. This gives those vertices degree k . Add a matching from the vertices in the smaller partite set to S . Now G is k -regular and has a Tutte set, so it has no 1-factor.

When any $k - 3$ vertices are deleted to form G' , what remains in each component H of $G - S$ is connected, due to the spanning biclique. Also some edge from H to S remains. If exactly one vertex of S remains, then G' is now connected. If more than one vertex of S remains, then any two are connected in G' by a path through some component of $G - S$.

3.3.19. *Every 3-regular simple graph with no cut-edge decomposes into copies of P_4 (the 4-vertex path).* By Petersen's Theorem, a 3-regular simple graph G with no cut-edge has a 1-factor M . Deleting the edges of M from G leaves a 2-factor H , which is a disjoint union of cycles. Choose an orientation D for the 2-factor so that each vertex has one edge in and one edge out (that is, make the components of the 2-factor into directed cycles).

We let each edge of M be the central edge in a copy of P_4 . The other two edges in the copy of P_4 containing $uv \in M$ are the edges leaving u and v in D ; let these be uw and vz . These three edges form P_4 if $w \neq z$. We have $w \neq z$ because each vertex has only one entering edge in D .

The central edges of these P_4 's are precisely the $n(G)/2$ edges of M . Each edge of H appears in exactly one of the constructed P_4 's, since every edge outside M follows exactly one vertex in D . Thus the copies of P_4 formed in this way are pairwise edge-disjoint and cover $E(G)$.

3.3.20. *A 3-regular simple graph G has a 1-factor if and only if it decomposes into copies of P_4 .*

Necessity. Deleting the edges of a 1-factor M from G leaves a 2-factor H , which is a disjoint union of cycles. Choose an orientation D for the 2-factor by choosing a consistent orientation around each cycle.

Let each edge of M be the central edge in a copy of P_4 . The other two edges in the copy of P_4 containing $uv \in M$ are the edges leaving u and v in D ; let these be uw and vz . These three edges form P_4 when $w \neq z$, which holds since each vertex has only one entering edge in D .

The central edges of these P_4 s are the $n(G)/2$ edges of M . Each edge of the 2-factor also appears in exactly one constructed P_4 , since each such edge is the tail of exactly one vertex in D . Thus these copies of P_4 are pairwise edge-disjoint and cover $E(G)$.

Sufficiency. A P_4 -decomposition of a 3-regular graph G has $n(G)/2$ subgraphs, since $e(G) = 3n(G)/2$ and $e(P_4) = 3$. No edge-disjoint copies of P_4 have a common internal vertex v , since that would give v degree at least 4. Hence the middle edges in the subgraphs of the decomposition form a matching of size $n(G)/2$ and hence a 1-factor.

3.3.21. *If G is a $2m$ -regular graph, and T is a tree with m edges and diameter less than the girth of G , then G decomposes into copies of T .* We prove a stronger result. Consider an arbitrary labeling of $V(T)$ with $\{1, \dots, m+1\}$. We prove by induction on m that G has a T -decomposition such that each vertex of G appears in $m+1$ copies of T , once with each label. Call this a *labeled T -decomposition* of G . The trivial necessary degree conditions are satisfied because the sum of the vertex degrees in T is $2m$. There will be $n(G)$ copies of T , independent of the value of m .

For $m = 0$ (or $m = 1$), the claim is immediate. For $m \geq 1$, let H be a 2-factor of G , and let i be a leaf of T , with neighbor j . Note that the distance in T from j to any other vertex of T is at most $\text{diam}(T) - 1$. The induction hypothesis guarantees a labeled $T - i$ -decomposition of $G - E(H)$. For each vertex w in each cycle in H , we add to the copy of $T - i$ with j at w by adding the edge to the next vertex in the cycle, which will then receive label i . This vertex does not already appear in this copy of $T - i$, because the girth of G exceeds the diameter of T .

3.3.22. *Hall's Theorem follows from Tutte's Theorem.* Given an X, Y -bigraph G , let H be the graph obtained from G by adding one vertex to Y if $n(G)$ is odd and then adding edges to turn Y into a clique.

a) *G has a matching of size $|X|$ if and only if H has a 1-factor.* Each edge of a matching in G has one vertex of X and one vertex of Y . Since $H[Y]$ is a clique, we can pair the remaining vertices arbitrarily to obtain a 1-factor in H from a matching of size $|X|$ in G . Conversely, if H has a 1-factor, it must use $|X|$ edges to saturate X , since $H[X]$ is an independent set. These edges from the desired matching in G .

b) *If G satisfies Hall's Condition ($|N(S)| \geq |S|$ for all $S \subseteq X$), then H satisfies Tutte's Condition ($o(H - T) \leq |T|$ for all $T \subseteq V(H)$).* Since $H[Y \cap T]$ is a clique, the odd components obtained by deleting T are the vertices of X whose neighbors all lie in T and perhaps the one large remaining component. Let $S = \{x \in X: N(x) \subseteq T \cap Y\}$. Since G satisfies Hall's Condition, $|S| \leq |T \cap Y| \leq |T|$. Thus $o(H - T) \leq |T| + 1$. Since $n(H)$ is even, $o(H - T)$ and $|T|$ have the same parity, and thus $o(H - T) \leq |T|$. Thus H satisfies Tutte's Condition.

c) *Tutte's Theorem implies Hall's Theorem.* The necessity of Hall's condition is immediate (any subset of X must have as many neighbors as elements to be completely matched). For sufficiency as a consequence of

Tutte's Theorem, we form H from G as described above. Since G satisfies Hall's Condition, part (b) implies that H satisfies Tutte's Condition. Tutte's Theorem now implies that H has a 1-factor. Part (a) now implies that G has a matching saturating X .

3.3.23. Consider a connected claw-free graph G of even order.

a) In a spanning tree of G generated by Breadth-First Search, any two vertices with a common parent other than the root are adjacent. Let r be the root, and let u, v be children of s , with $s \neq r$. In a Breadth-First Search tree, the path from the root to any vertex is a shortest such path. We have $d(u, r) = d(v, r) = d(s, r) + 1$. An edge from u or v to the parent t of s would establish a shorter path to the root of length $d(s, r)$. Hence there is no such edge. To avoid having s, t, u, v induce a claw, u and v must be adjacent.

b) G has a 1-factor. We use induction on $n(G)$. Basis step ($n(G) = 2$): G must be K_2 .

Induction step ($n(G) > 2$): Let T be a Breadth-First Search tree from r . Let u be a vertex at maximum distance from r , and let s be the parent of u . If s has no other child, then $T' = T - \{u, s\}$ is connected. If s has another child v , then let $T' = T - \{u, v\}$.

In each case, T' is connected, so $G'[V(T')]$ is a smaller connected claw-free graph of even order. The induction hypothesis guarantees a perfect matching in G' . To this matching we add the edge between the two vertices we deleted to obtain G' (in the first case, the edge us exists because s is the parent of u ; in the second case, the edge uv exists by part (a).)

3.3.24. Maximum number of edges with no 1-factor. A maximal n -vertex graph with no 1-factor consists of m vertices of degree $n - 1$, with the remaining $n - m$ vertices inducing a union of $m + 2$ cliques of odd order. Since adding an edge cannot reduce α' or increase it by more than one, we may assume that $\alpha'(G) = n/2 - 1$. Hence $\max_{S \subseteq V} \{o(G - S) - |S|\} = 2$; the maximum matching omits 2 vertices. Let S be a set achieving equality, so $o(G - S) = |S| + 2$. Each component of $G - S$ must induce a clique, $G - S$ has no component of even order (else add edges from even to odd components), and vertices of S have degree $n - 1$, all because adding the edges that would be missing if any of these failed would not reduce the deficiency of S . This completely describes the maximal graphs.

The maximum number of edges in a graph with minimum degree $k < n/6 - 2$ and no 1-factor is $\binom{k}{2} + k(n - k) + \binom{n - 2k - 1}{2}$. We assume n is even. Let G be a maximal n -vertex graph with no 1-factor. Let $f(k) = \binom{k}{2} + k(n - k) + \binom{n - 2k - 1}{2}$. We first observe that there is a maximal graph having no 1-factor that has $f(k)$ edges and minimum degree k ; the graph is $K_k \vee ((k + 1)K_1 + K_{n - 2k - 1})$. This is an example of the structure above with $m = k$. This construction is valid when $n - 2k - 1 > 0$, which requires only

$k \leq n/2 - 1$. Nevertheless, $f(k)$ is not always the maximum size of a graph H with minimum degree k and no 1-factor. For $n \geq 8$, we can build counterexamples when $k < n/2 - 1$ and k is at least $n/6$ (approximately). The smallest counterexample occurs when $n = 8$ and $k = 2$. We have $f(2) = 16$ and $f(3) = 18$. We obtain a graph with minimum degree 2, no 1-factor, and 17 edges by deleting one edge from $K_3 \vee (5K_1)$.

Suppose H has a vertex of degree $k < n/6 - 2$ and has no perfect matching; we obtain an upper bound on $e(H)$. Augment H by adding edges to obtain a maximal supergraph G having no 1-factor; note that $\delta(G) = l \geq k$. By direct computation, $f(t + 1) - f(t) = 3t - n + 4$. If $t \geq n/2$, then there is no graph with minimum degree at least t that has no 1-factor. Since $f(t)$ is a parabola centered at $t = (n - 4)/3$ and $\delta(G) < n/2$, we have $f(k) \geq f(l)$ if $k < n/6 - 2$. Therefore, it suffices to prove that if $\delta(G) = k$ and G has the form described above, then $e(G) \leq f(k)$.

If v does not have degree $n - 1$, then v belongs to a clique of size $d(v) - m + 1$ in $G - S$, which is odd. If any two components of $G - S$ have sizes $p \geq q \geq k - m + 3$, then we gain $2p$ and lose $2q - 4$ edges by moving two vertices from the smaller to the larger clique in $G - S$, still maintaining the same minimum degree. Hence for fixed m , $e(G)$ is maximized by using cliques of size $k - m + 1$ for all but one component of $G - S$. Now the degree sum in G is $m(n - 1) + (k - m + 1)(m + 1)k + [n - (k - m + 1)(m + 1)][n - (k - m + 1)(m + 1)]$. If $m < k/2$, then replacing m by $k - m$ increases this, since only the first term changes. Hence we may assume $m \geq k/2$. If $k > m \geq k/2 - 1$, then we can increase m by moving two vertices from a small clique to S , since $2(m + 1) \geq 2(k - m - 1)$ guarantees that we can then take vertices from the other small components to make new components of size $k - m - 1$. This increases the number of edges (computation omitted), so we may assume $m = k$. Now $e(G) = f(k)$.

3.3.25. A graph G is factor-critical if and only if $n(G)$ is odd and $o(G - S) \leq |S|$ for all nonempty $S \subseteq V(G)$. Necessity. Factor-critical graphs are those where every subgraph obtained by deleting one vertex has a 1-factor. Thus factor-critical graphs have odd order. Given a nonempty subset S of $V(G)$, let v be a member of S , and let $G' = G - v$. Since G' has a 1-factor and $G - S = G' - S'$, we have $o(G - S) = o(G' - S') \leq |S'| = |S| - 1$. Thus the desired inequality holds for S .

Sufficiency. Suppose that $n(G)$ is odd and $o(G - S) \leq |S|$ for all nonempty $S \subseteq V(G)$. Given a vertex $v \in V(G)$, let $G' = G - v$. Consider $S' \subseteq V(G')$, and let $S = S' \cup \{v\}$. Since G has odd order, the quantities $|S|$ and $o(G - S)$ have different parity. The hypothesis thus yields $o(G - S) \leq |S| - 1$. Since $G' - S' = G - S$, we have $o(G' - S') = o(G - S) \leq |S| - 1 = |S'|$. By Tutte's Theorem, $G - v$ has a 1-factor.

3.3.26. If M is a matching in a graph G , and u is an M -unsaturated vertex, and G has no M -augmenting path that starts at u , then u is unsaturated in some maximum matching in G . We use induction on the difference between $|M|$ and $\alpha'(G)$. If the difference is 0, then already u is unsaturated in some maximum matching. If M is not a maximum matching, then there is an M -augmenting path P . Since u is not an endpoint of P , then $M \Delta E(P)$ is a larger matching M' that does not saturate u .

If no M' -augmenting path starts at u , then by the induction hypothesis u is unsaturated in some maximum matching. Suppose that an M' -augmenting path P' starts at u . Since no M -augmenting path starts at u , P' shares an edge with P . Form an M -alternating path by following P' from u until it first reaches a vertex of P ; this uses no edges of $M' - M$. Then follow P to the end in whichever direction continues the M -alternating path. Since both endpoints of P are M -unsaturated, this completes an M -augmenting path starting at u , which contradicts the hypothesis.

3.3.27. Proof of Tutte's 1-Factor Theorem from correctness of the Blossom Algorithm.

a) If G has no perfect matching, and M is a maximum matching in G , and S and T are the sets generated when running the Blossom Algorithm from u , then $|T| < |S| \leq o(G - T)$. Since M is a maximum matching, no M -augmenting path is found. At the start of the algorithm, $|T| = 0 < 1 = |S|$. As the algorithm proceeds, exploring a vertex $v \in S$ leads to consideration of a neighbor y of v that is not in T . If y is unsaturated in M , then an M -augmenting path is found; this case is forbidden by hypothesis. If y is saturated but not yet in S , then y is added to T and its mate is added to S . This augments both T and S by one element, maintaining $|S| = |T| + 1$.

Finally, if y is saturated and lies in S , then a blossom is established. Every matched edge in the blossom has one vertex of T and one vertex of S , and the blossom is shrunk into the single vertex of S on it that is not matched to another vertex along the blossom. Hence the "current" S and T shrink by the same number of vertices. Therefore, we always maintain $|S| = |T| + 1$.

Now consider the second inequality. When the algorithm ends, the current S is an independent set with no edges to vertices that have not been reached, because if an edge has been found between vertices of S , then a blossom would have been shrunk, and an edge to an unreached vertex would have been explored. Hence deleting T leaves at least $|S|$ odd components. (There may be more among the unreached vertices; the algorithm does not explore unsaturated edges from T .)

These isolated vertices in the final S correspond to odd components of $G - T$ in the original graph, because shrinking of a blossom loses an

even number of vertices. Since all edges leaving the blossom are explored, the final T disconnects everything in the blossom from what is outside of it. Since also the blossom corresponds to an odd number of vertices in the original graph, deleting T from the original graph must leave an odd component among these vertices. One such set exists for each vertex in the final S (at the end of the algorithm).

b) Proof of Tutte's 1-Factor Theorem. If $o(G - T) \leq |T|$ for every vertex subset T , then the algorithm cannot end by finding a set T such that $|T| < |S| \leq o(G - T)$. Hence it can only end by finding an augmentation, and this continues until a 1-factor is found.

3.3.28. a) Reduction of the f -factor problem to the f -solubility problem. It suffices to prove that G has an f -factor if and only if the graph H obtained by replacing each edge by a path of length 3 is f' -soluble, where f' is the extension of f obtained by defining f' to equal 1 on all the new vertices.

Suppose that x, a, b, y is the path in H representing the edge xy in G . If G has an f -factor, then H is f' -soluble by giving weights 1, 0, 1 or 0, 1, 0 to the successive edges on the path depending on whether xy is or is not in the f -factor.

Conversely, if H is f' -soluble, then because every edge of H is incident to a vertex with $f' = 1$, every edge is used with weight 1 or 0, and the weights along the 3-edge path representing an edge xy must be 1, 0, 1 or 0, 1, 0. At a vertex x there must be exactly $f(x)$ paths of the first type, so we obtain an f -factor of G by using the edges corresponding to these paths.

b) Reduction of f -solubility to 1-factor. Let H be the graph formed from G by replacing each vertex $v \in V(G)$ with an independent set of $f(v)$ vertices. Now G is f -soluble if and only if H has a 1-factor; collapsing or expanding the vertices turns the solution of one problem into the solution of the other.

3.3.29. Tutte's f -factor condition and graphic sequences. For disjoint sets Q, T , let $e(Q, T)$ denote the number of edges from Q to T . For a function h defined on $V(G)$, let $h(S) = \sum_{v \in S} h(v)$ for $S \subseteq V(G)$.

For $f: V(G) \rightarrow \mathbb{N}_0$, Tutte proved that G has an f -factor if and only if

$$q(S, T) + f(T) - d_{G-S}(T) \leq f(S)$$

for all choices of disjoint subsets $S, T \subseteq V(G)$, where $q(S, T)$ is the number of components Q of $G - S - T$ such that $e(Q, T) + f(V(Q))$ is odd. a) **The Parity Lemma.** The quantity $\delta(S, T)$ has the same parity as $f(V)$ for disjoint sets $S, T \subseteq V(G)$, where $\delta(S, T) = f(S) - f(T) + d_{G-S}(T) - q(S, T)$. We use the observation that the parity of the number of odd values in a set of integers equals the parity of the sum of the set.

We fix S and use induction on $|T|$. When $T = \emptyset$, we have $f(T) - d_{G-S}(T) = 0$. Also $m(Q, T) = 0$ for each component Q of $G - S - T$, so we can sum over the components to obtain $q(S, \emptyset) \equiv f(\bar{S}) \pmod{2}$. Signs don't matter, so $\delta(S, \emptyset) \equiv f(\bar{S}) - f(S) \equiv f(V(G)) \pmod{2}$.

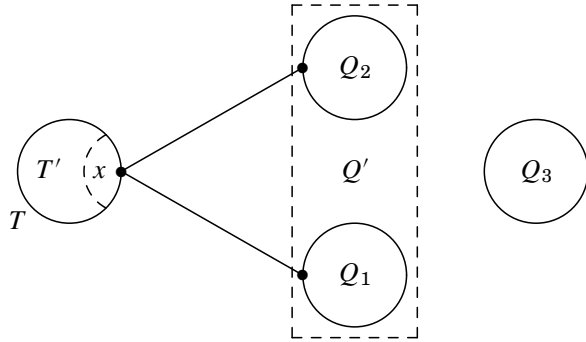
For $T \neq \emptyset$, we compare $\delta(S, T)$ and $\delta(S, T - x)$; it suffices to show that the difference is even. Let $T' = T - x$. In computing the difference, the contributions of $-f(S)$ cancel, as do the sums over T' . This leaves

$$\delta(S, T) - \delta(S, T') = q(S, T) - q(S, T') + f(x) - d_{G-S}(x).$$

The contributions to $q(S, T)$ and $q(S, T')$ from components of $G - S - T$ having no neighbors of x also cancel. The components having neighbors of x combine with x to form one large component in $G - S - T'$ with vertex set $Q' \cup \{x\}$. Our initial observation about parity yields

$$q(S, T) - q(S, T') \equiv f(Q') + m(Q', T) - [f(Q' \cup \{x\}) + m(Q' \cup \{x\}, T')] \pmod{2}.$$

The edges from x to Q' count in $m(Q', T)$, the edges from x to T' count in $m(Q' \cup \{x\}, T')$, and the edges from Q' to T' count in both. Thus $m(Q', T) - m(Q' \cup \{x\}, T') \equiv d_{G-S}(x) \pmod{2}$, and we have $q(S, T) - q(S, T') \equiv f(x) + d_{G-S}(x) \pmod{2}$. This has the same parity as the rest of the difference, yielding $\delta(S, T) - \delta(S, T') \equiv 0 \pmod{2}$.



b) Let $d_1, \dots, d_n$ be nonnegative integers with $\sum d_i$ even and $d_1 \geq \dots \geq d_n$. If $G = K_n$ and $f(v_i) = d_i$, then G has an f -factor if and only if $\sum_{i=1}^k d_i \leq (n-1-s)k + \sum_{i=n+1-s}^n d_i$ for all k, s with $k+s \leq n$. Such an f -factor exists if and only if $f(V) = \sum d_i$ is even and Tutte's condition holds. Since $d_{K_n-S}(v) = n-1-|S|$ for all $v \in T$, the f -factor condition requires that $f(T) \leq f(S) + (n-1-|S|)|T| - q(S, T)$ for any disjoint sets $S, T \subseteq V(G)$. With $|T| = k$ and $|S| = s$, the condition becomes $f(T) \leq f(S) + (n-1-s)k - q(S, T)$.

Necessity. Applied with the k vertices of largest degree in T and the s

vertices of smallest degree in S and using $q(S, T) \geq 0$, the f -factor condition yields the desired inequality.

Sufficiency. It suffices to establish inequality shown above to be equivalent to the f -factor condition. Since $K_n - S - T$ is connected, always $q(S, T) \leq 1$. Since $\sum f(v_i)$ is even, the two sides of the inequality have the same parity, by the Parity Lemma. It therefore suffices to prove that $f(T) \leq f(S) + (n-1-s)k$ when S and T are disjoint. With $s = |S|$ and $k = |T|$, it suffices to prove the inequality when T consists of the k vertices of largest degree and S consists of the s vertices of smallest degree. It then becomes the given inequality for $d_1, \dots, d_n$.

c) Nonnegative integers $d_1, \dots, d_n$ with $d_1 \geq \dots \geq d_n$ are the vertex degrees of a simple graph if and only if $\sum d_i$ is even and $\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min\{k, d_i\}$ for $1 \leq k \leq n$. Any realization can be viewed as an f -factor of K_n , where $f(v_i) = d_i$. Thus it suffices to show that this condition is equivalent to the condition in part (b).

For fixed $|S|, |T|$, this inequality is always satisfied if and only if it is satisfied when $T = \{x_i, \dots, x_k\}$ and $S = \{x_{n+1-s}, \dots, x_n\}$, in which case it becomes $\sum_{i=1}^k f_i \leq \sum_{i=n+1-s}^n f_i + (n-1-s)k = (n-1)k + \sum_{i=n+1-s}^n (f_i - k)$. This is always satisfied if and only if it is satisfied when the right side attains its minimum over $0 \leq s \leq n-k$, which happens when $n+1-s = \min\{i: i > k \text{ and } f_i < k\}$. Since $(n-1)k = k(k-1) + (n-k)k$, the value is then $k(k-1) + \sum_{i=k+1}^n \min\{k, f_i\}$.